Group 22 Midterm Project

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1. Data preprocessing

Load packages and data

```r
library(tidyverse)
library(GGally)
library(corrplot)
library(caret)
library(glmnet)
library(mgcv)
library(earth)
library(pls)
library(patchwork)
library(vip)
```

```r
load("datA.RData")
dfA <- datA
```

Basic Structure

```r
# Dimensions
dim(dfA)
```

```
## [1] 2000   17
```

```r
# Summary
summary(dfA)
```

```
##       id                 age          menopause           race
##  Length:2000        Min.   :25.00   Min.   :0.0000   Min.   :1.000
##  Class :character   1st Qu.:47.00   1st Qu.:0.0000   1st Qu.:1.000
##  Mode  :character   Median :55.00   Median :1.0000   Median :1.000
##                     Mean   :54.99   Mean   :0.6555   Mean   :1.697
##                     3rd Qu.:62.40   3rd Qu.:1.0000   3rd Qu.:2.000
##                     Max.   :85.00   Max.   :1.0000   Max.   :4.000
##       bmi          tumor_size     tumor_grade     lymph_nodes
##  Min.   :16.00   Min.   :0.300   Min.   :1.000   Min.   : 0.000
##  1st Qu.:23.00   1st Qu.:2.100   1st Qu.:2.000   1st Qu.: 0.750
##  Median :26.05   Median :3.100   Median :2.000   Median : 2.000
##  Mean   :26.06   Mean   :3.384   Mean   :2.182   Mean   : 2.623
##  3rd Qu.:29.10   3rd Qu.:4.800   3rd Qu.:3.000   3rd Qu.: 4.000
##  Max.   :42.60   Max.   :8.300   Max.   :3.000   Max.   :16.000
##        ER               PR              HER2            Ki67
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.000   Min.   : 1.00
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:13.50
##  Median :1.0000   Median :0.0000   Median :0.000   Median :21.60
##  Mean   :0.6405   Mean   :0.4805   Mean   :0.222   Mean   :21.85
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:29.90
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.000   Max.   :62.30
```

```
##      stage         surgery          chemo        hormonal_therapy
##  Min.   :1.00   Min.   :0.000   Min.   :0.0000   Min.   :0.000
##  1st Qu.:1.00   1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.000
##  Median :2.00   Median :1.000   Median :1.0000   Median :0.000
##  Mean   :2.05   Mean   :0.507   Mean   :0.5525   Mean   :0.467
##  3rd Qu.:3.00   3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.000
##  Max.   :3.00   Max.   :1.000   Max.   :1.0000   Max.   :1.000
##     log_PFS
##  Min.   :-0.5345
##  1st Qu.: 1.6724
##  Median : 2.1911
##  Mean   : 2.1957
##  3rd Qu.: 2.7125
##  Max.   : 5.1013
```

Convert variables to factors

```r
dfA <- dfA |>
  mutate(
    # Nominal variables
    menopause = factor(menopause, levels = c(0,1), labels = c("Pre","Post")),
    race = factor(race, levels = c(1,2,3,4),
                  labels = c("White","Black","Asian","Hispanic")),
    surgery = factor(surgery, levels = c(0,1),
                     labels = c("BCS","Mastectomy")),
    chemo = factor(chemo, levels = c(0,1),
                   labels = c("No","Yes")),
    hormonal_therapy = factor(hormonal_therapy, levels = c(0,1),
                              labels = c("No","Yes")),
    ER = factor(ER, levels = c(0,1), labels = c("Negative","Positive")),
    PR = factor(PR, levels = c(0,1), labels = c("Negative","Positive")),
    HER2 = factor(HER2, levels = c(0,1), labels = c("Negative","Positive")),

    # Ordinal variables
    tumor_grade = factor(tumor_grade, levels = c(1,2,3),
                         labels = c("Well","Moderate","Poor"),
                         ordered = TRUE),

    stage = factor(stage, levels = c(1,2,3),
                   labels = c("Stage I","Stage II","Stage III"),
                   ordered = TRUE)
  )
```

Missing data check

```r
# Count
colSums(is.na(dfA))
```

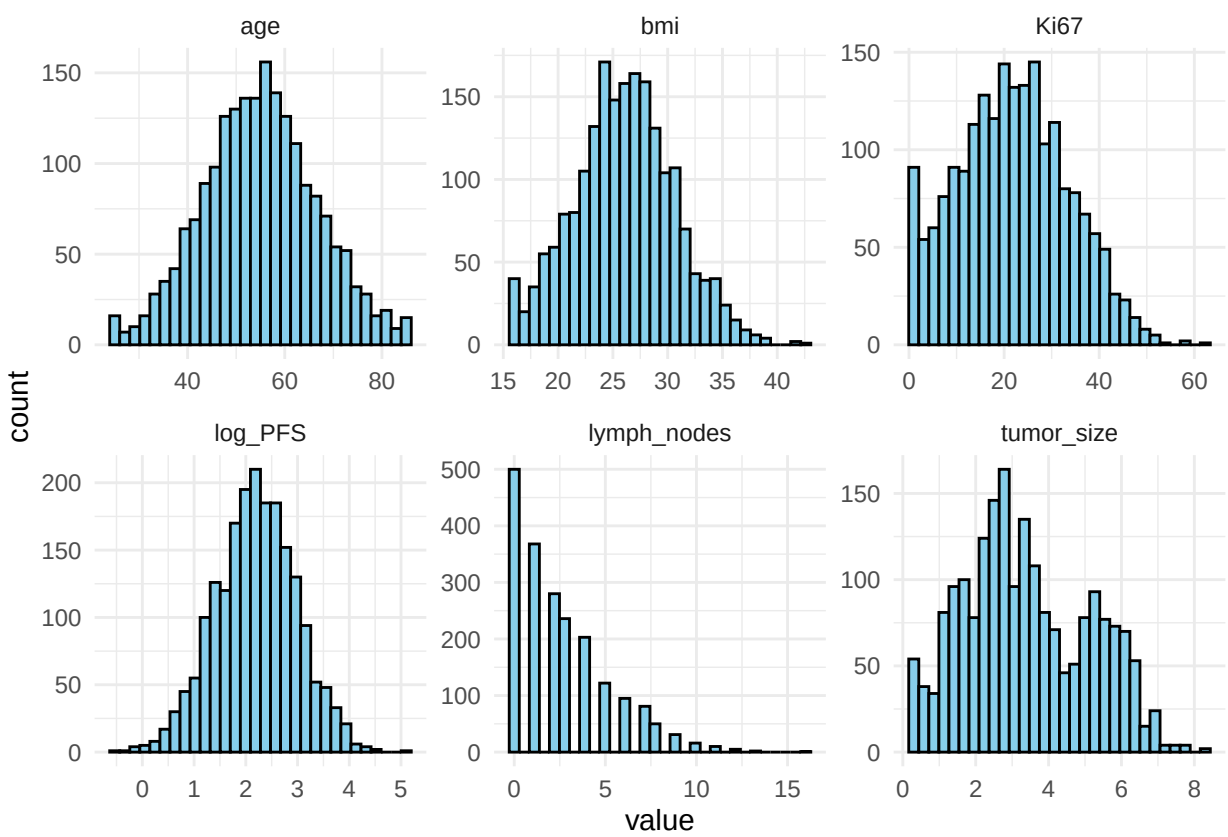
```
##                id               age         menopause              race
##                 0                 0                 0                 0
```

```
##                bmi       tumor_size      tumor_grade      lymph_nodes
##                  0                0                0                0
##                 ER               PR             HER2             Ki67
##                  0                0                0                0
##              stage          surgery            chemo hormonal_therapy
##                  0                0                0                0
##           log_PFS
##                  0
```

2. EDA for Dataset A

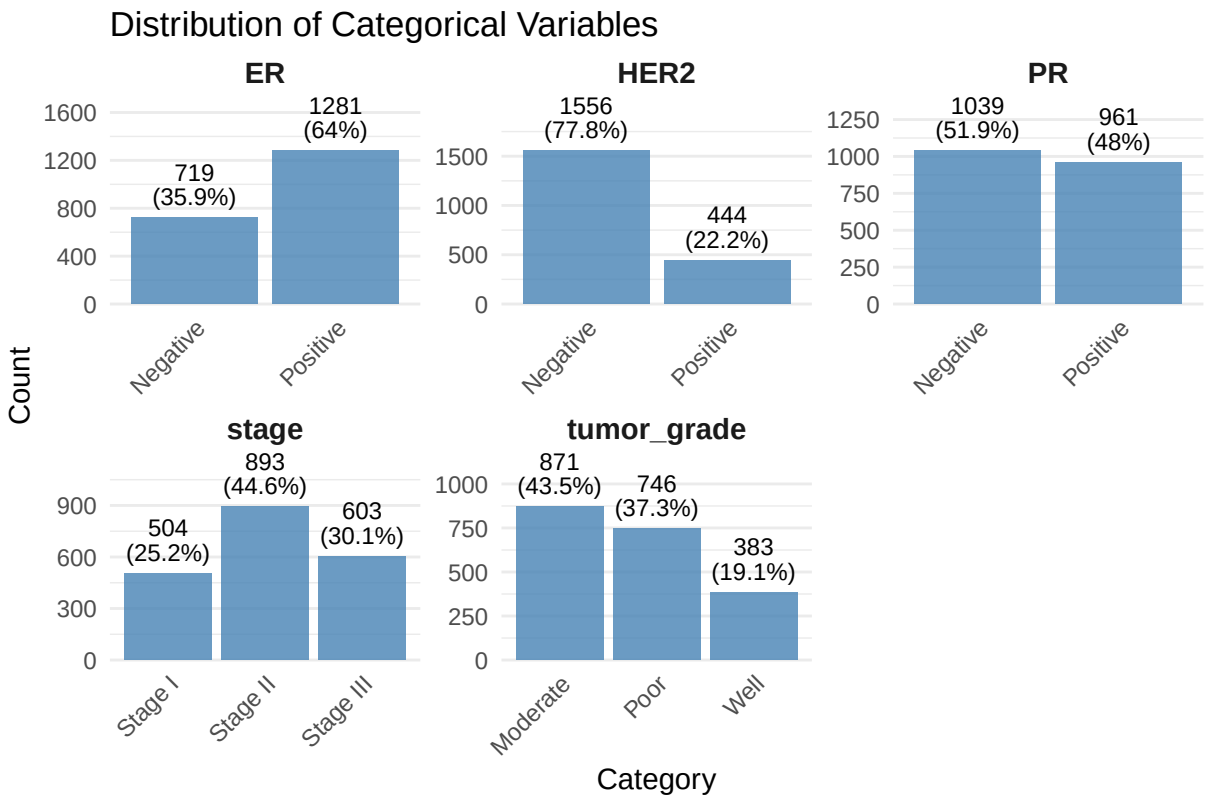
Distribution of continuous variables

```
# Histograms
dfA |>
  select(age, bmi, tumor_size, lymph_nodes, Ki67, log_PFS) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  facet_wrap(~name, scales = "free") +
  theme_minimal()
```

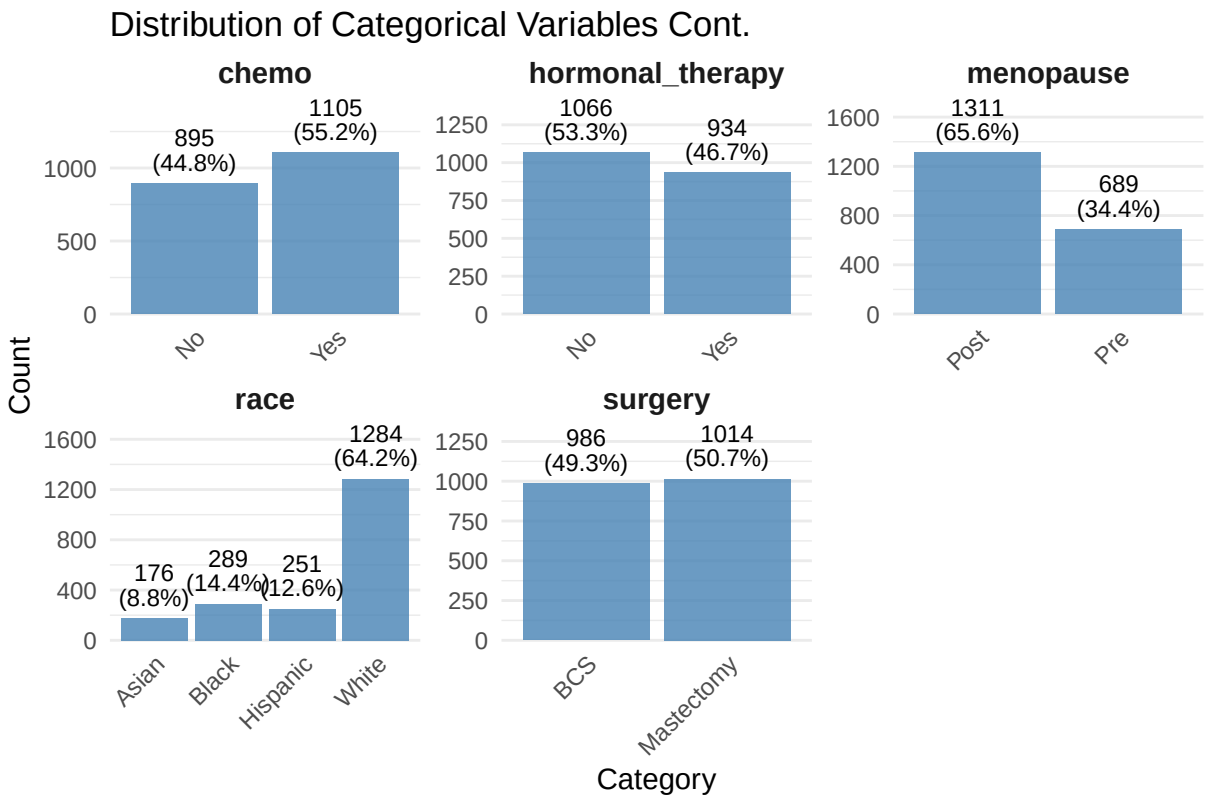


Frequency distribution of categorical variables

```r
plot_categorical <- function(data, vars, title_text) {
  plot_data <- data |>
    select(all_of(vars)) |>
    mutate(across(everything(), as.character)) |>
    pivot_longer(cols = everything()) |>
    group_by(name, value) |>
    summarise(count = n(), .groups = "drop") |>
    group_by(name) |>
    mutate(
      percent = count / sum(count),
      label = paste0(count, "\n(", round(percent * 100, 1), "%)")
    )

  ggplot(plot_data, aes(x = value, y = count)) +
    geom_bar(stat = "identity", fill = "steelblue", alpha = 0.8) +
    geom_text(
      aes(label = label),
      vjust = -0.3,
      color = "black",
      size = 3,
      lineheight = 0.9
    ) +
    facet_wrap(~name, scales = "free", ncol = 3) +
    scale_y_continuous(expand = expansion(mult = c(0, 0.25))) +
    coord_cartesian(clip = "off") +
    theme_minimal() +
    labs(x = "Category", y = "Count", title = title_text) +
    theme(
      strip.text = element_text(face = "bold", size = 11, margin = margin(b = 10)),
      axis.text.x = element_text(angle = 45, hjust = 1),
      panel.grid.major.x = element_blank(),
      plot.margin = margin(t = 20, r = 10, b = 10, l = 10),
      legend.position = "none"
    )
}

vars1 <- c("stage", "tumor_grade", "ER", "PR", "HER2")
vars2 <- c("race", "menopause", "chemo", "hormonal_therapy", "surgery")

plot_categorical(dfA, vars1, "Distribution of Categorical Variables")
```

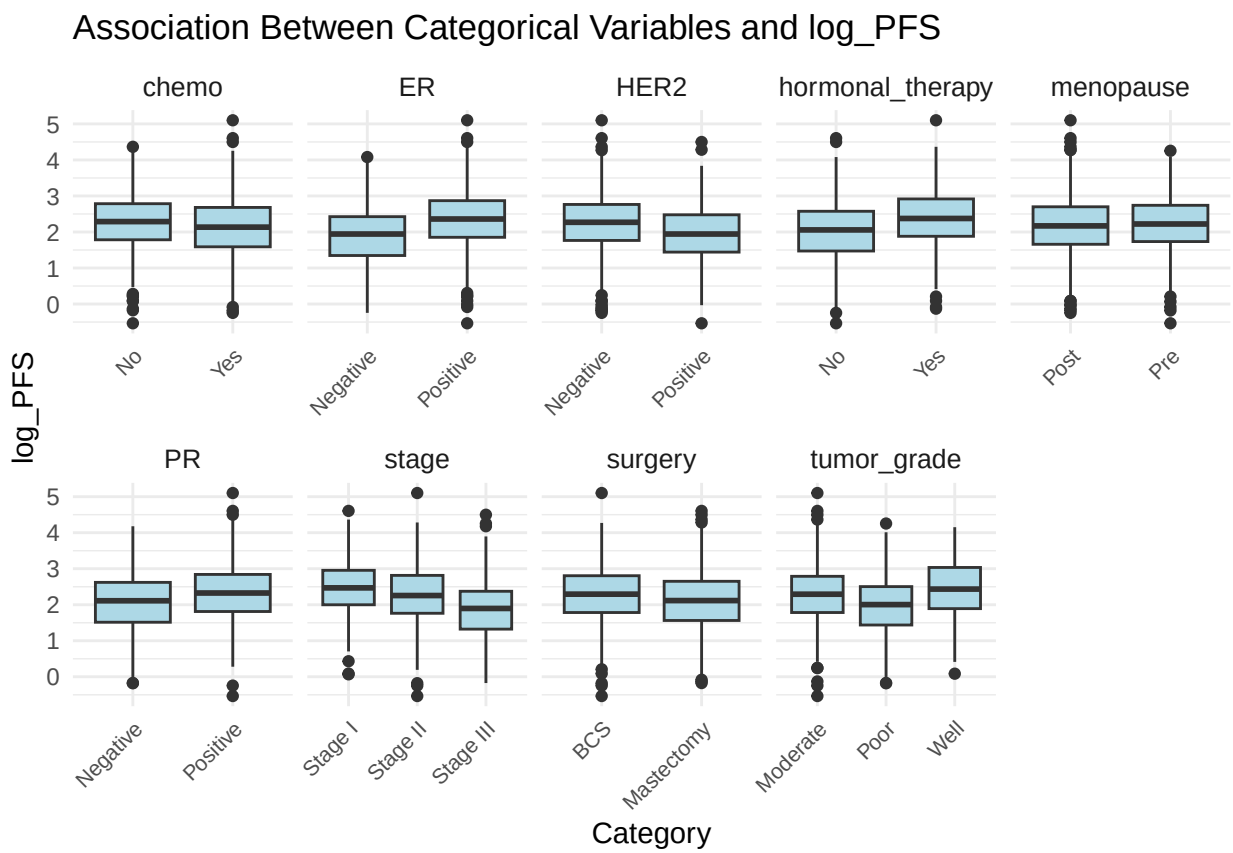



```
plot_categorical(dfA, vars2, "Distribution of Categorical Variables Cont.")
```



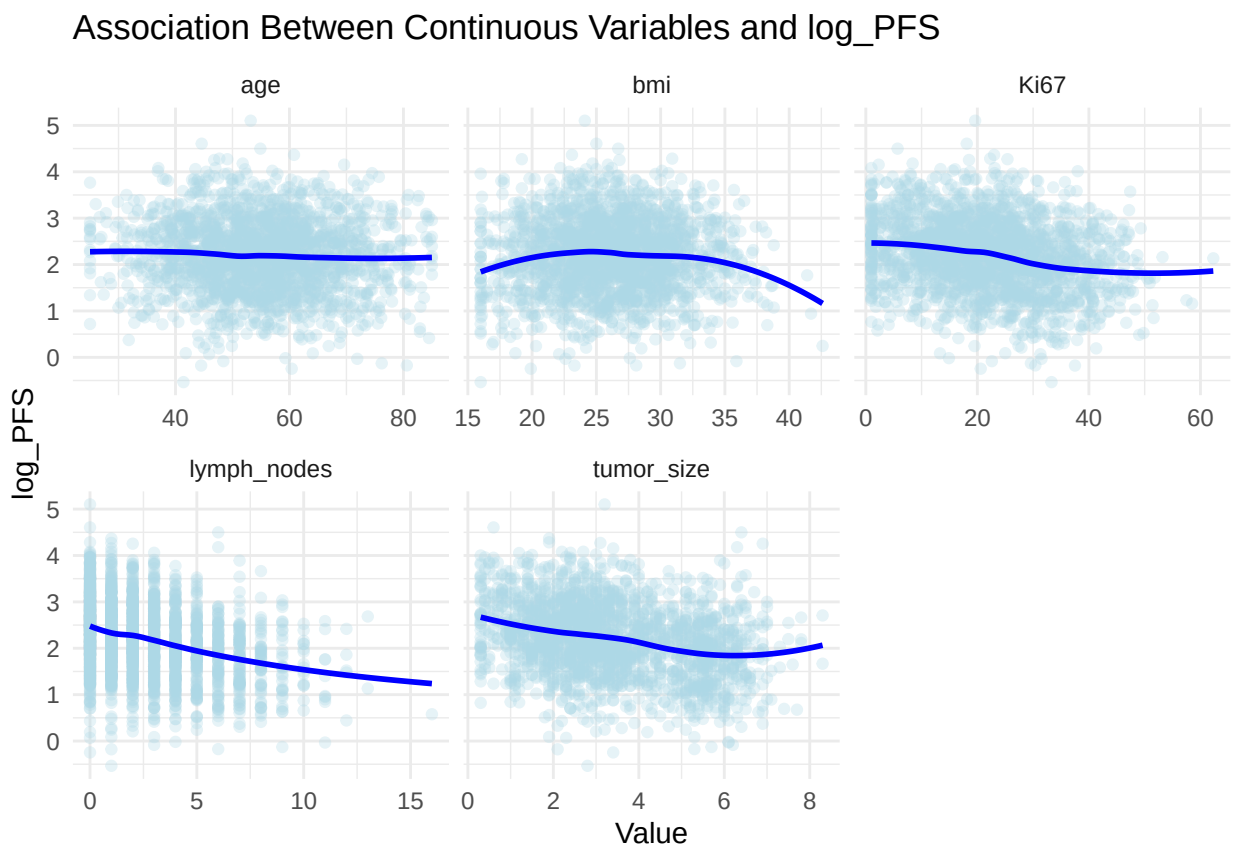
Relationship with response variable

```
# Categorical
dfA |>
  select(log_PFS, stage, tumor_grade, ER, PR, HER2,
         chemo, hormonal_therapy, surgery, menopause) |>
  mutate(across(-log_PFS, as.character)) |>
  pivot_longer(cols = -log_PFS) |>
  ggplot(aes(x = value, y = log_PFS)) +

  geom_boxplot(fill = "lightblue") +

  facet_wrap(~name, scales = "free_x", ncol = 5) +

  theme_minimal() +
  labs(
    x = "Category",
    y = "log_PFS",
    title = "Association Between Categorical Variables and log_PFS"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(size = 10)
  )
```



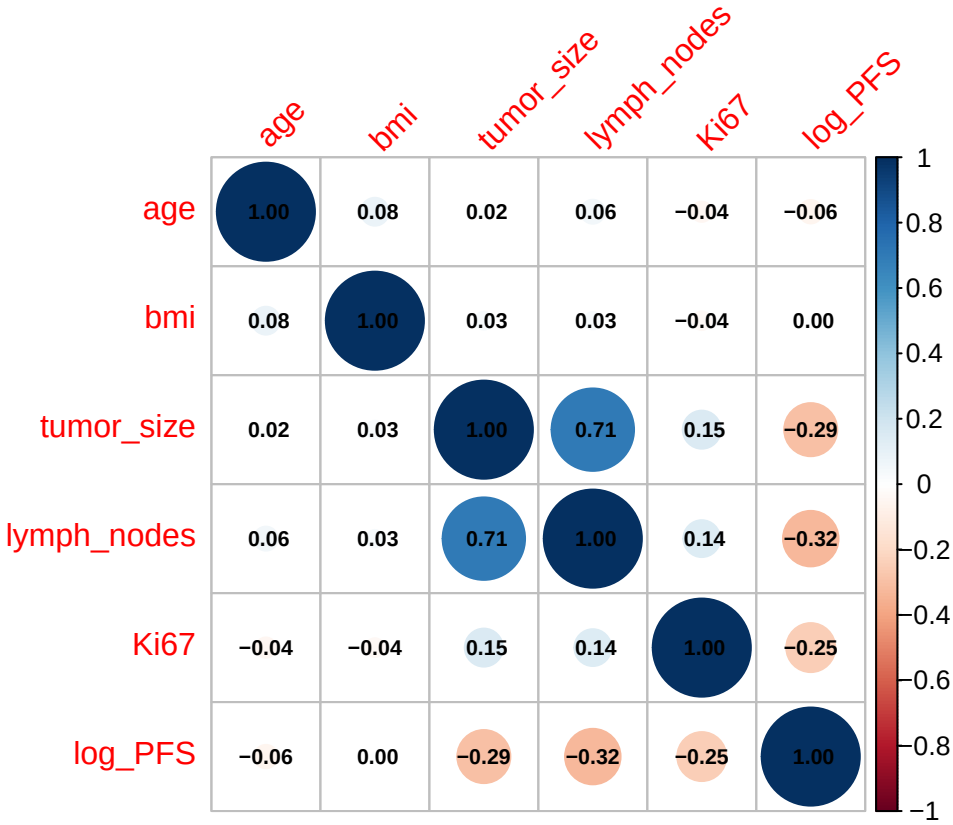

```r
# Continuous
dfA |>
  select(log_PFS, age, bmi, tumor_size, lymph_nodes, Ki67) |>
  pivot_longer(cols = -log_PFS) |>
  ggplot(aes(x = value, y = log_PFS)) +
  geom_point(alpha = 0.3, color = "lightblue") +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  facet_wrap(~name, scales = "free_x") +
  theme_minimal() +
  labs(
    x = "Value",
    y = "log_PFS",
    title = "Association Between Continuous Variables and log_PFS"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



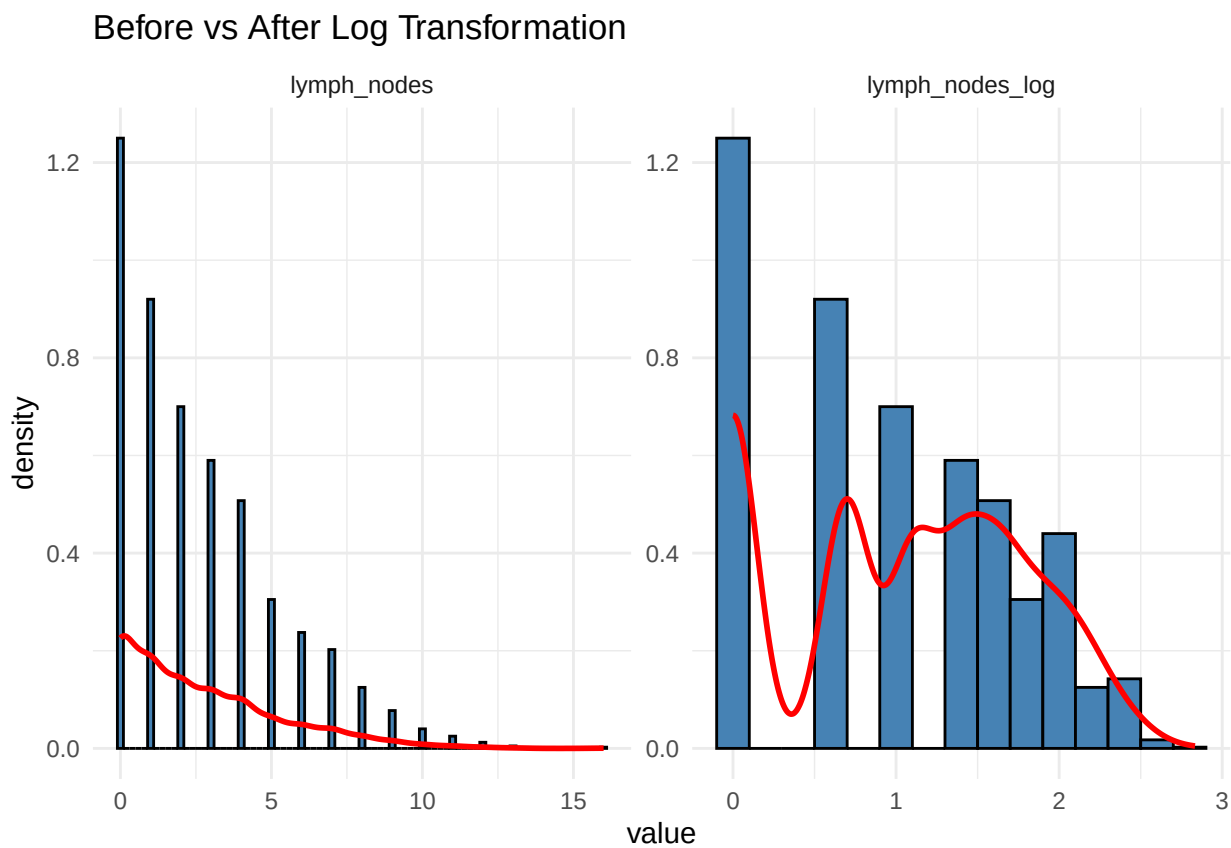
Correlation Check (Continuous)

```
library(corrplot)
cont_vars <- dfA |>
  select(age, bmi, tumor_size, lymph_nodes, Ki67, log_PFS)

cor_matrix <- cor(cont_vars, use = "complete.obs")

corrplot::corrplot(
  cor_matrix,
  method = "circle",
  type = "full",
  tl.col = "red",
  tl.srt = 45,
  addCoef.col = "black",
  number.cex = 0.7,
  mar = c(0,0,1,0)
)
```



Handle Skewness

```
# Log transform lymph_nodes
dfA <- dfA |>
  mutate(
    lymph_nodes_log = log(lymph_nodes + 1)
  )
dfA |>
  select(lymph_nodes, lymph_nodes_log) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(aes(y = after_stat(density)),
                 binwidth = 0.2,
                 fill = "steelblue",
                 color = "black") +
  geom_density(color = "red", linewidth = 1) +
  facet_wrap(~name, scales = "free") +
  theme_minimal() +
  labs(title = "Before vs After Log Transformation")
```



Train Test Split

```
set.seed(1)

train_index <- createDataPartition(dfA$log_PFS, p = 0.7, list = FALSE)
train_data <- dfA[train_index, ]
test_data  <- dfA[-train_index, ]
```

We loaded the cleaned dataset from clean_datA.RData, which had already been split into a training set (train_data, n = 1400) and a testing set (test_data, n = 600). To ensure reproducibility and match with previous data cleaning process, we set the random seed to 1 before model fitting.

```
set.seed(1)
load("clean_datA.RData")
train <- train_data
test <- test_data
```

All model development and tuning were performed exclusively within the training set using 10-fold cross-validation.

```
#cross validation setting
ctrl <- trainControl(method = "cv", number = 10, selectionFunction = "best")
```

We first created a variable to include all predictors as well as the "lymph_nodes_log" we created.

```
#full list of predictors
full_formula <- log_PFS ~ age + menopause + race + bmi + tumor_size + tumor_grade +
  lymph_nodes_log + ER + PR + HER2 + Ki67 + stage + surgery + chemo + hormonal_therapy
```

3. Model Building

3-1. Linear Regression (Baseline)

The baseline linear regression model was fit using the full set of predictors. In the baseline model, older age, larger tumor size, greater lymph node burden, HER2 positivity, and lack of hormonal therapy were associated with shorter log-transformed PFS, whereas ER positivity was associated with longer log-transformed PFS.

```
set.seed(1)
fit_lm <- train(full_formula, data = train, method = "lm", trControl = ctrl)
fit_lm$finalModel
```

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Coefficients:
##         (Intercept)                 age        menopausePost
##           2.852e+00          -5.174e-03           -2.218e-05
##           raceBlack           raceAsian         raceHispanic
##          -1.628e-02           5.358e-02           -1.896e-02
##                 bmi          tumor_size        tumor_grade.L
##           1.163e-03          -8.058e-02           -8.738e-02
##       tumor_grade.Q      lymph_nodes_log           ERPositive
##          -3.051e-02          -2.350e-01            2.269e-01
##          PRPositive         HER2Positive                 Ki67
##          -4.304e-03          -1.560e-01           -3.974e-03
##             stage.L              stage.Q    surgeryMastectomy
##           1.324e-01          -5.666e-02            2.536e-02
##            chemoYes  hormonal_therapyYes
##           2.228e-02           1.507e-01
```

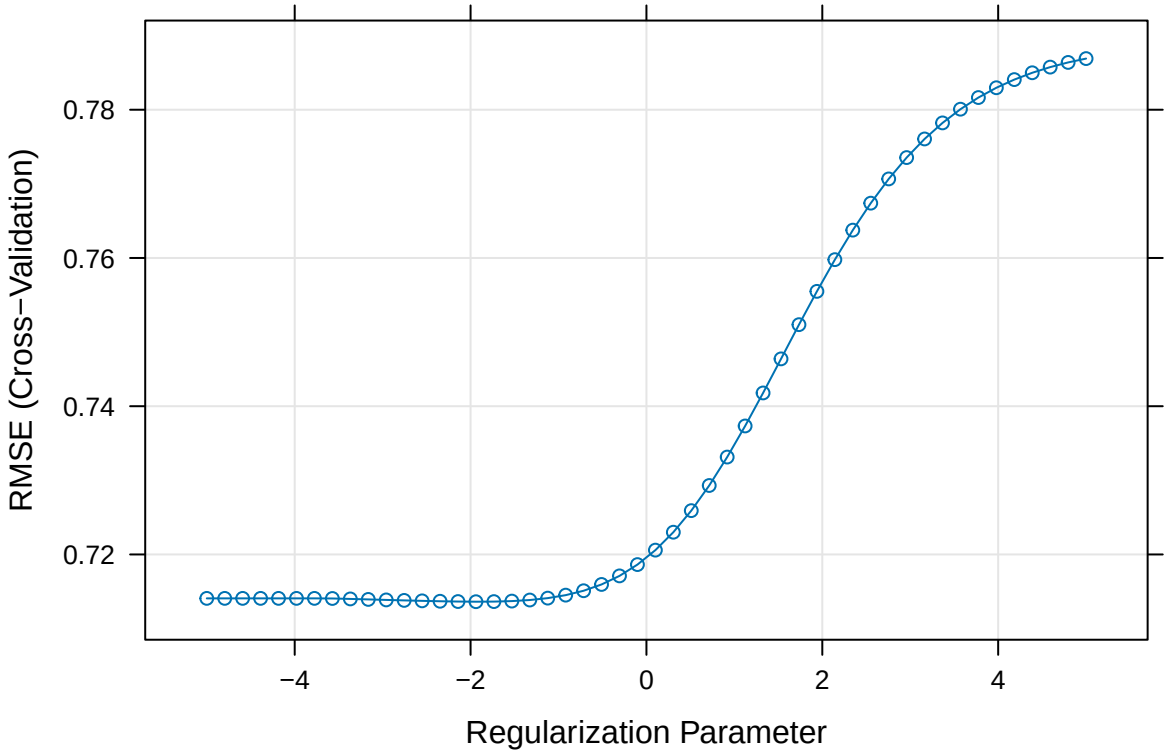
3-2. Ridge, Lasso, Elastic Net (Regularization)

3-2-1. Ridge Regression

The ridge regression was fit using the full set of predictors with alpha = 0 based on definition. The ridge regression model will choose the best lambda value by itself.

```
set.seed(1)
fit_ridge <- train(full_formula, data = train, method = "glmnet",
                   tuneGrid = expand.grid(alpha = 0,
                                          lambda = exp(seq(-5, 5, length = 50))),
                   trControl = ctrl)
```

```r
plot(fit_ridge, xTrans = log)
```




```r
fit_ridge$bestTune
```

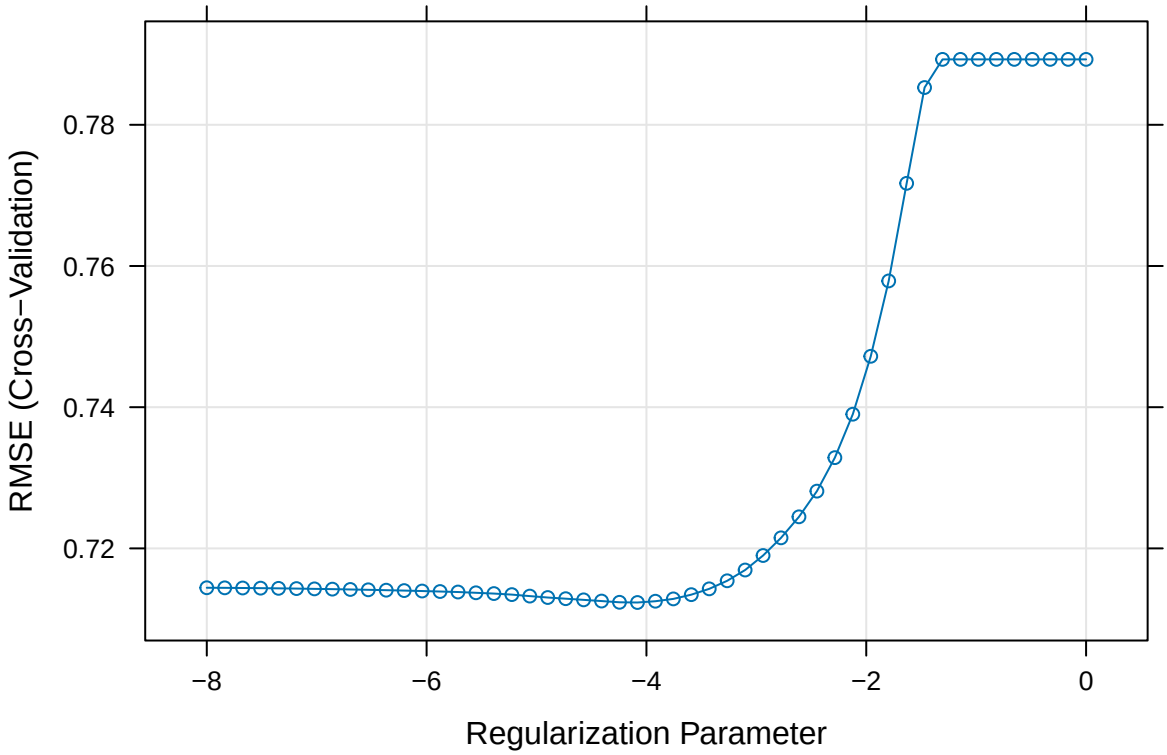
```
##    alpha  lambda
## 16     0 0.14388
```

From the ridge regression plot, we can see that the performance of CV RMSE was pretty well with about 0.71 as the lowest.

3-2-2. Lasso Regression

The lasso regression was fit using the full set of predictors with alpha = 1 based on definition. The lasso regression model will choose the best lamda value by itself.

```r
set.seed(1)
fit_lasso <- train(full_formula, data = train, method = "glmnet",
                   tuneGrid = expand.grid(alpha = 1,
                                          lambda = exp(seq(-8, 0, length = 50))),
                   trControl = ctrl)

plot(fit_lasso, xTrans = log)
```




```r
fit_lasso$bestTune
```

```
##    alpha     lambda
## 25     1 0.01687988
```

```r
coef(fit_lasso$finalModel, fit_lasso$bestTune$lambda)
```

```
## 20 x 1 sparse Matrix of class "dgCMatrix"
##                      s=0.01687988
## (Intercept)           2.693881472
## age                  -0.003818717
## menopausePost         .
## raceBlack             .
## raceAsian             0.001121165
## raceHispanic          .
## bmi                   .
## tumor_size           -0.050594800
## tumor_grade.L        -0.059838793
## tumor_grade.Q        -0.004566234
## lymph_nodes_log      -0.188641673
## ERPositive            0.225909932
## PRPositive            .
## HER2Positive         -0.120831687
## Ki67                 -0.003964691
## stage.L               .
```

```
## stage.Q              -0.040577531
## surgeryMastectomy     .
## chemoYes              .
## hormonal_therapyYes  0.124811922
```

Moreover, from the result curve of lasso regression model, we can observe that there is a obvious downwards trend when the tuning parameter approaches to -4. The cv rmse is slightly smaller than 0.71 at parameter = -4.

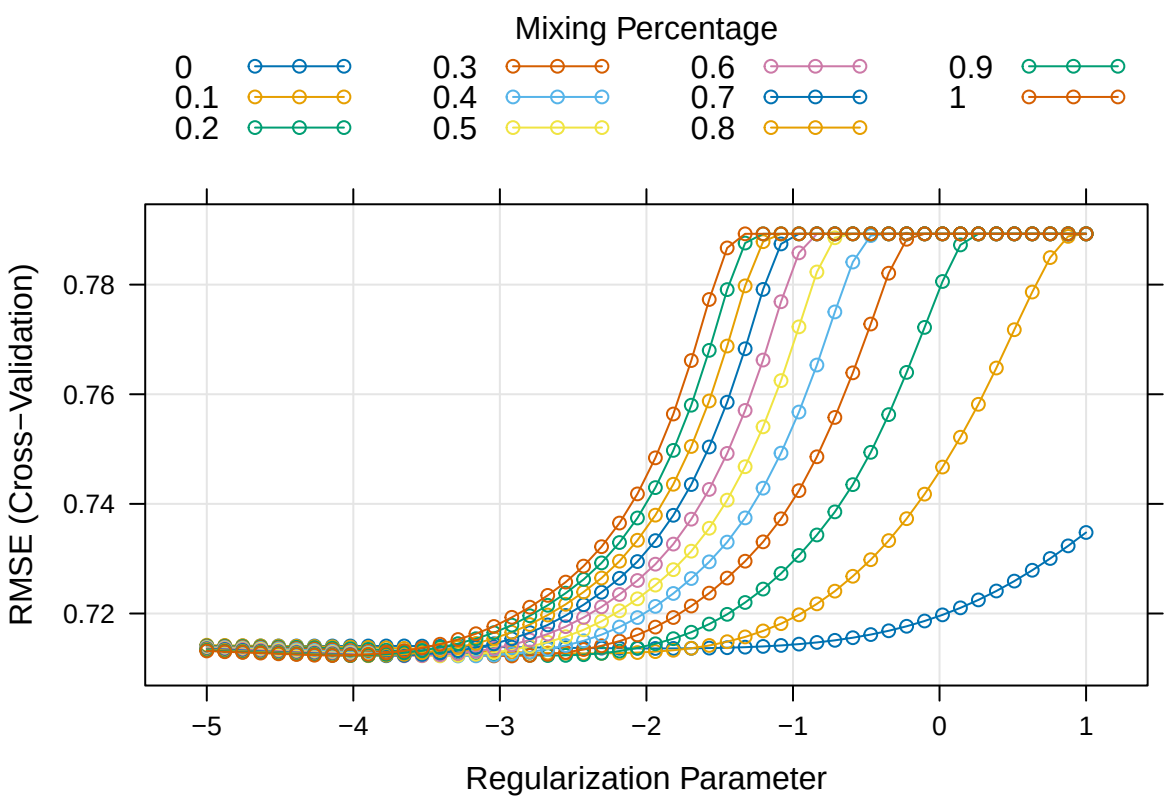
3-2-3. Elastic net Regression

The elastic net regression was fit using the full set of predictors without choosen alpha. The elastic net regression model will choose the best lamda value and alpha value by itself.

```r
set.seed(1)
fit_enet <- train(full_formula, data = train, method = "glmnet",
                  tuneGrid = expand.grid(alpha = seq(0, 1, length = 11),
                                         lambda = exp(seq(-5, 1, length = 50))),
                  trControl = ctrl)

fit_enet$bestTune
```

```
##     alpha     lambda
## 168   0.3 0.05402181
```

```r
plot(fit_enet, xTrans = log)
```



```r
coef(fit_enet$finalModel, fit_enet$bestTune$lambda)
```

```
## 20 x 1 sparse Matrix of class "dgCMatrix"
##                      s=0.05402181
## (Intercept)           2.689200530
## age                  -0.003737587
## menopausePost         .
## raceBlack             .
## raceAsian             0.004031234
## raceHispanic          .
## bmi                   .
## tumor_size           -0.051318471
## tumor_grade.L        -0.061140619
## tumor_grade.Q        -0.006597589
## lymph_nodes_log      -0.179525096
## ERPositive            0.215327564
## PRPositive            .
## HER2Positive         -0.118858793
## Ki67                 -0.004032789
## stage.L               .
## stage.Q              -0.040508838
## surgeryMastectomy     .
## chemoYes              .
## hormonal_therapyYes   0.126287877
```

The output and the plot shows that when alpha = 0.3 and lambda = 0.05402181, the lowest cv rmse value is about 0.71

3-3. PCR & PLS (Dimension Reduction)

3-3-1. PCR (Principal Component Regression)

The PCR model was fit using the full set of predictors with standardized variables. It reduced the original predictors to a smaller number of principal components and selected the number of components by cross-validation.

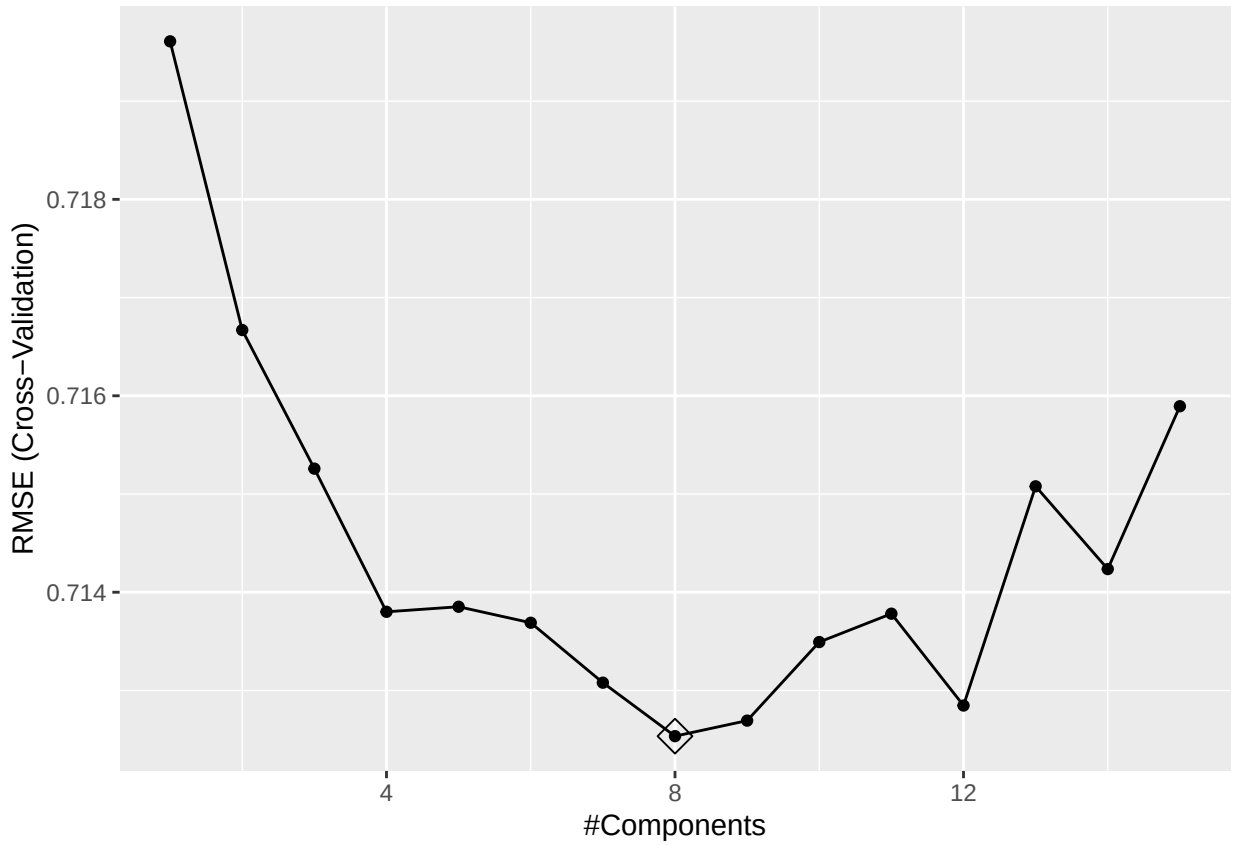
```r
set.seed(1)
pcr_mod <- pcr(full_formula, data = train, scale = TRUE, validation = "CV")

summary(pcr_mod)
```

```
## Data:    X dimension: 1400 19
##  Y dimension: 1400 1
## Fit method: svdpc
## Number of components considered: 19
##
## VALIDATION: RMSEP
## Cross-validated using 10 random segments.
##        (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
## CV          0.7904   0.7215   0.7187   0.7171   0.7157   0.7162   0.7167
## adjCV       0.7904   0.7214   0.7186   0.7170   0.7156   0.7160   0.7165
```

```
##        7 comps  8 comps  9 comps  10 comps  11 comps  12 comps  13 comps
## CV      0.7159   0.7153   0.7158    0.7160    0.7166    0.7163    0.7168
## adjCV   0.7156   0.7150   0.7156    0.7157    0.7163    0.7160    0.7164
##        14 comps  15 comps  16 comps  17 comps  18 comps  19 comps
## CV       0.7161    0.7165    0.7155    0.7165    0.7168    0.7153
## adjCV    0.7156    0.7160    0.7150    0.7159    0.7162    0.7147
##
## TRAINING: % variance explained
##          1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps  8 comps
## X          17.10    29.62    38.79    45.68    51.73    57.68    63.23    68.40
## log_PFS    16.81    17.58    18.05    18.42    18.43    18.55    18.72    18.94
##          9 comps  10 comps  11 comps  12 comps  13 comps  14 comps  15 comps
## X          73.33     77.93     82.36     86.52     90.35     93.57     95.27
## log_PFS    18.96     19.06     19.12     19.34     19.36     19.68     19.68
##          16 comps  17 comps  18 comps  19 comps
## X           96.84     98.26     99.50    100.00
## log_PFS     19.93     20.01     20.06     20.53
```

```r
fit_pcr <- train(full_formula, data = train_data, method = "pcr",
                 scale = TRUE, trControl = ctrl, tuneLength = 15)
ggplot(fit_pcr, highlight = TRUE)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the caret package.
##   Please report the issue at <https://github.com/topepo/caret/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

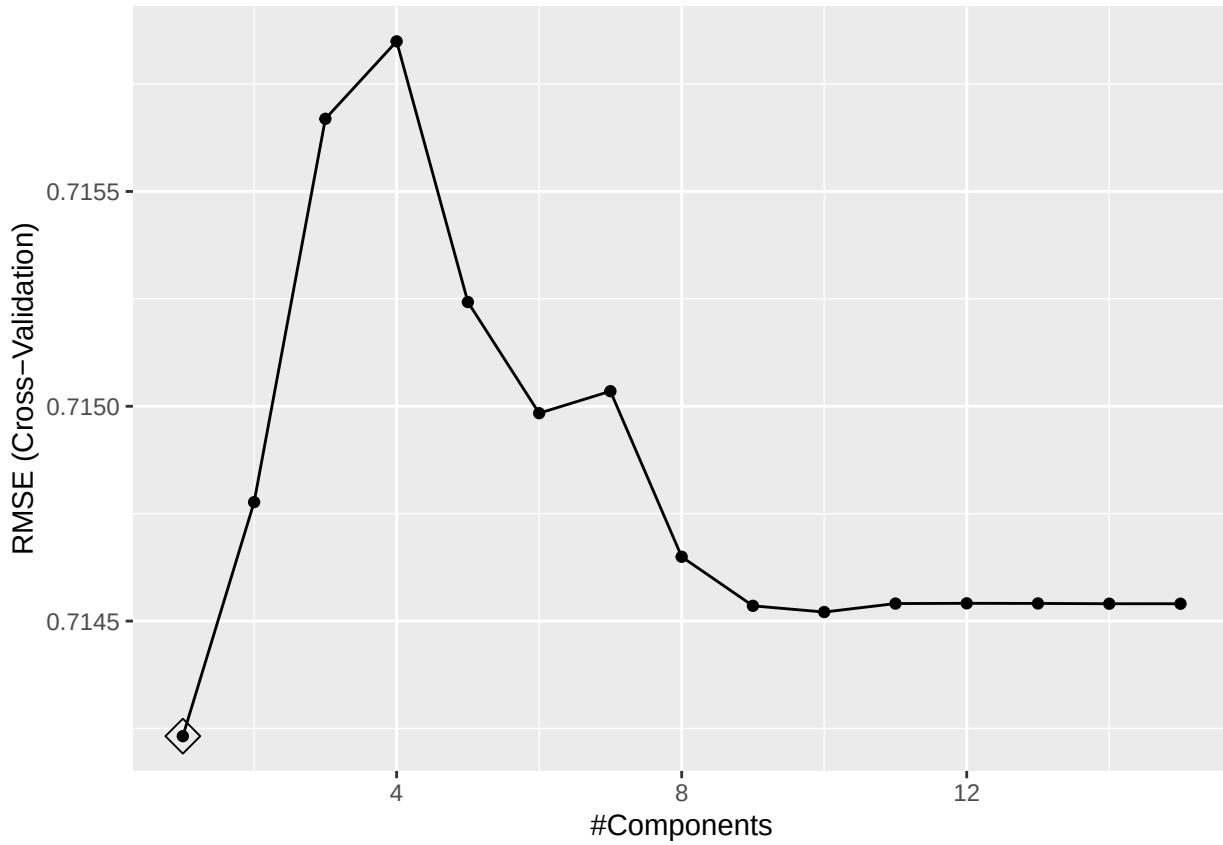


The PCR tuning plot suggests that predictive performance improved as the number of components increased up to about 8, where the CV RMSE reached its minimum, about 0.71. Beyond that point, adding more components did not improve prediction.

3-3-2. PLS (Partial Least Squares)

The PLS model was also fit using the full set of standardized predictors, with the number of components selected by cross-validation. PLS constructs components using information from both the predictors and the response.

```
set.seed(1)
fit_pls <- train(full_formula, data = train, method = "pls",
                 scale = TRUE, trControl = ctrl, tuneLength = 15)
ggplot(fit_pls, highlight = TRUE)
```



From the plot, we can observe that PLS reached its best performance with only 1 component. This suggests that PLS was more efficient in extracting components that were directly related to the outcome.

3-4. GAM & MARS (Nonlinear Methods)

3-4-1. GAM

The GAM model was fit using the full set of predictors to allow for possible nonlinear effects.

```
set.seed(1)
fit_gam <- train(full_formula, data = train, method = "gam", trControl = ctrl)

summary(fit_gam)
```

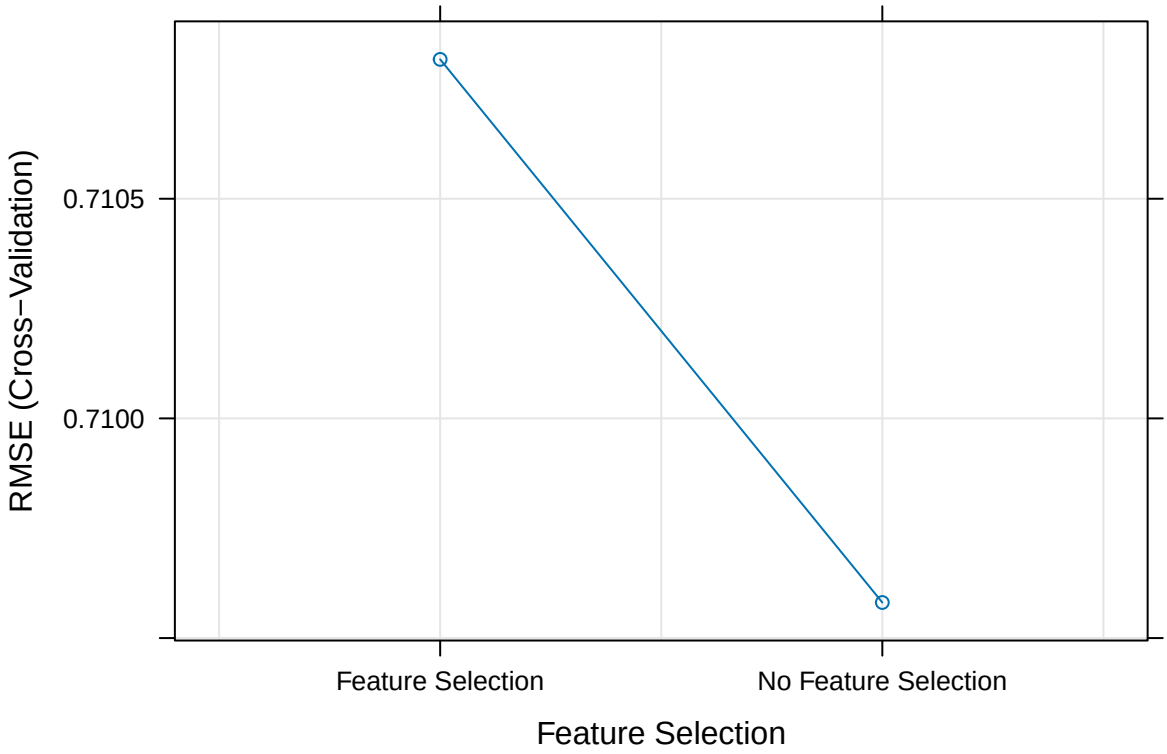
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## .outcome ~ menopausePost + raceBlack + raceAsian + raceHispanic +
##     tumor_grade.Q + ERPositive + PRPositive + HER2Positive +
##     stage.Q + surgeryMastectomy + chemoYes + hormonal_therapyYes +
##     tumor_grade.L + stage.L + s(lymph_nodes_log) + s(tumor_size) +
##     s(bmi) + s(Ki67) + s(age)
##
```

```
## Parametric coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.9868323  0.0612685  32.428  < 2e-16 ***
## menopausePost        0.0008707  0.0563421   0.015 0.987673
## raceBlack           -0.0031062  0.0563060  -0.055 0.956014
## raceAsian            0.0682970  0.0655041   1.043 0.297299
## raceHispanic        -0.0234697  0.0589238  -0.398 0.690466
## tumor_grade.Q       -0.0287353  0.0320250  -0.897 0.369728
## ERPositive           0.2193459  0.0629101   3.487 0.000505 ***
## PRPositive           0.0031413  0.0471963   0.067 0.946944
## HER2Positive        -0.1505532  0.0461769  -3.260 0.001140 **
## stage.Q             -0.0866823  0.0434926  -1.993 0.046455 *
## surgeryMastectomy    0.0351675  0.0400235   0.879 0.379733
## chemoYes             0.0148415  0.0390448   0.380 0.703919
## hormonal_therapyYes  0.1493363  0.0482501   3.095 0.002008 **
## tumor_grade.L       -0.0883196  0.0506801  -1.743 0.081612 .
## stage.L              0.1240411  0.0941050   1.318 0.187685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                      edf Ref.df      F  p-value
## s(lymph_nodes_log) 2.237  2.791 10.996 1.68e-06 ***
## s(tumor_size)      2.252  2.897  5.071  0.00249 **
## s(bmi)             3.503  4.398  6.369 2.97e-05 ***
## s(Ki67)            1.000  1.000  2.990  0.08403 .
## s(age)             1.000  1.000  4.953  0.02621 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.215   Deviance explained = 22.8%
## GCV = 0.49907  Scale est. = 0.49016   n = 1400
```

```r
fit_gam$finalModel
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## .outcome ~ menopausePost + raceBlack + raceAsian + raceHispanic +
##     tumor_grade.Q + ERPositive + PRPositive + HER2Positive +
##     stage.Q + surgeryMastectomy + chemoYes + hormonal_therapyYes +
##     tumor_grade.L + stage.L + s(lymph_nodes_log) + s(tumor_size) +
##     s(bmi) + s(Ki67) + s(age)
##
## Estimated degrees of freedom:
## 2.24 2.25 3.50 1.00 1.00  total = 24.99
##
## GCV score: 0.4990726
```

```r
plot(fit_gam)
```

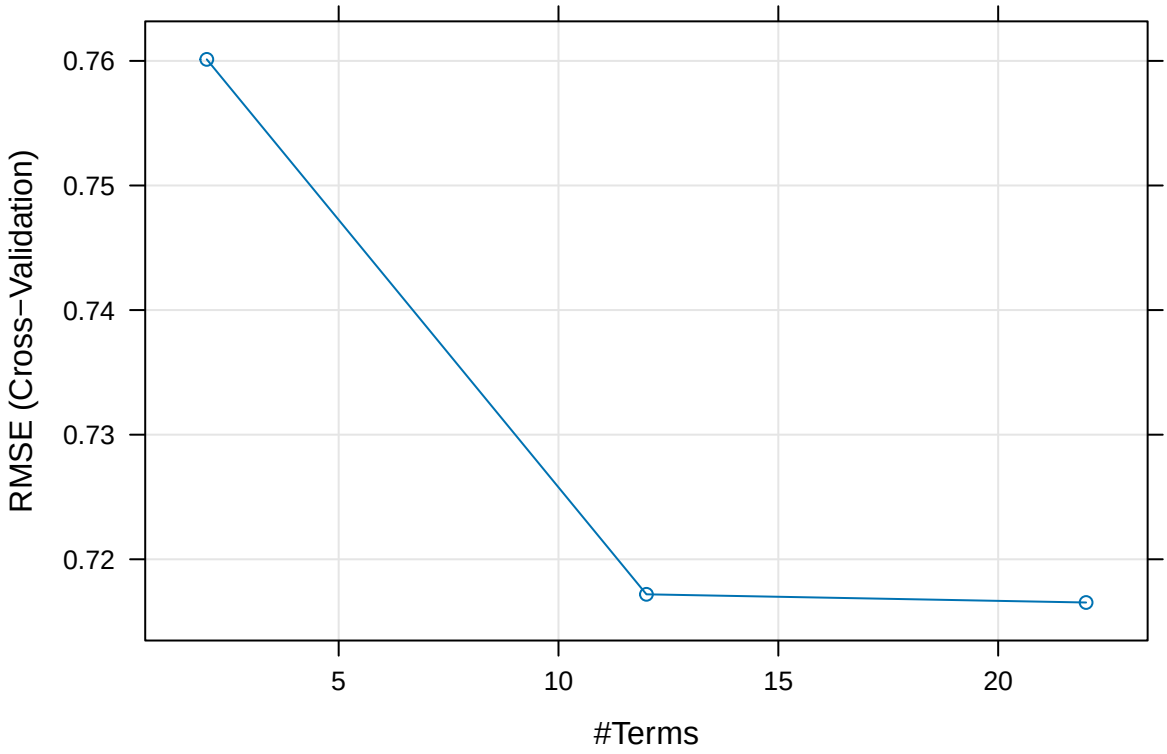


From the plot, we can see that the cv rmse was slightly smaller for the GAM without feature selection, indicating that keeping all predictors produced better predictive performance than using feature selection.

3-4-2. MARS

The MARS model was fit as a flexible nonlinear regression model using the full set of predictors. It automatically searched for piecewise linear relationships and selected model complexity through cross-validation.

```
set.seed(1)
fit_mars <- train(full_formula, data = train, method = "earth", trControl = ctrl)

plot(fit_mars)
```



From the result, it shows that although the minimum CV RMSE was achieved at the largest number of terms, the difference from the model with 12 terms was negligible. This suggests that increasing model complexity beyond 12 terms did not meaningfully improve prediction.

4. Model comparison using CV-RMSE

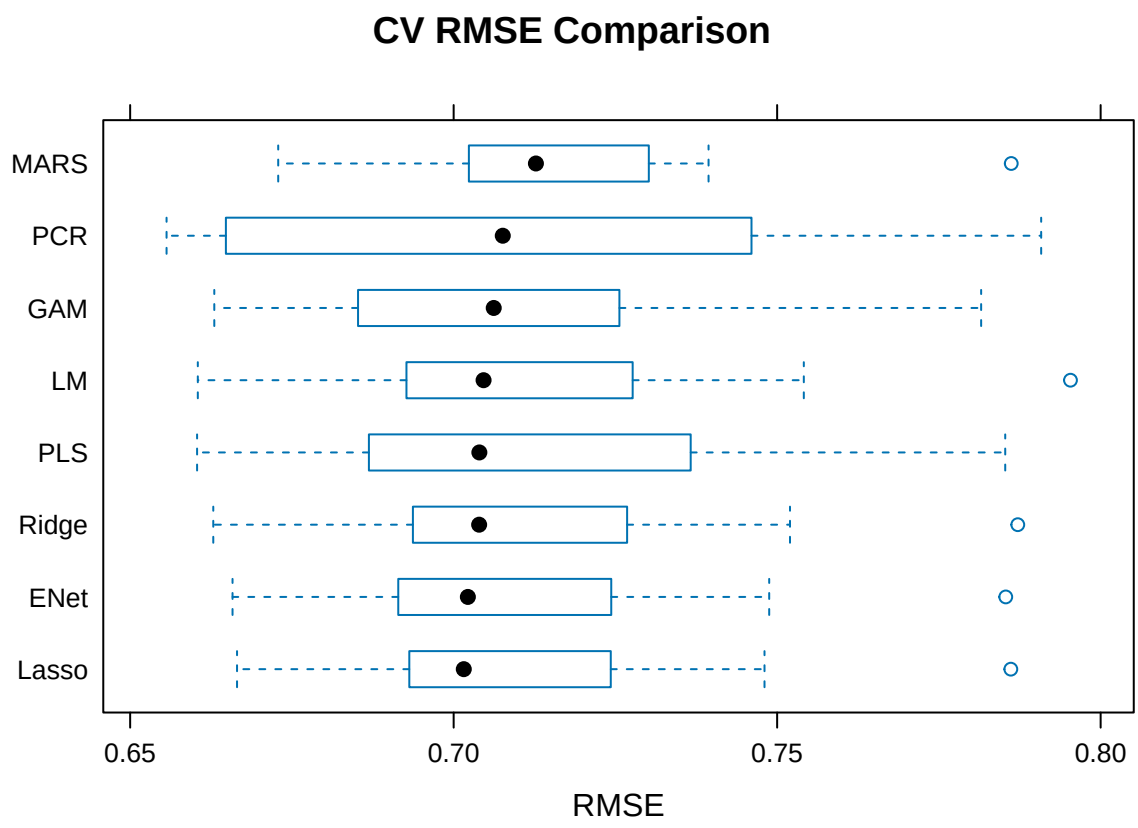
Based on the cross-validated RMSE results, the GAM achieved the smallest mean RMSE among all candidate models. This indicates that, during model development within the training set, the GAM had the strongest overall predictive performance. Since cross-validation provides an internal estimate of out-of-sample prediction error, this result suggests that the GAM was the most promising model among the candidate approaches we considered.

```r
# obtain all cv rmse results from the models
res_comparison <- resamples(list(LM = fit_lm, Ridge = fit_ridge, Lasso = fit_lasso,
                                 ENet = fit_enet, PLS = fit_pls, PCR = fit_pcr,
                                 GAM = fit_gam, MARS = fit_mars))
summary(res_comparison)
```

```
##
## Call:
## summary.resamples(object = res_comparison)
##
## Models: LM, Ridge, Lasso, ENet, PLS, PCR, GAM, MARS
## Number of resamples: 10
##
```

```
## MAE
##            Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.5337108 0.5522548 0.5624079 0.5679073 0.5768032 0.6353839    0
## Ridge 0.5371991 0.5557381 0.5624472 0.5678412 0.5740277 0.6267160    0
## Lasso 0.5367243 0.5534992 0.5620403 0.5656592 0.5718693 0.6241170    0
## ENet  0.5358611 0.5526348 0.5626724 0.5656471 0.5719386 0.6233326    0
## PLS   0.5413678 0.5560137 0.5664547 0.5685936 0.5772124 0.6238078    0
## PCR   0.5304338 0.5394183 0.5616313 0.5663350 0.5850971 0.6179636    0
## GAM   0.5366209 0.5546442 0.5635412 0.5664443 0.5701414 0.6177419    0
## MARS  0.5470988 0.5581213 0.5702772 0.5718815 0.5781167 0.6252596    0
##
## RMSE
##            Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.6604551 0.6936360 0.7046218 0.7145404 0.7274328 0.7953390    0
## Ridge 0.6628472 0.6948594 0.7039394 0.7136240 0.7265387 0.7871919    0
## Lasso 0.6665183 0.6939369 0.7015638 0.7123377 0.7242293 0.7861174    0
## ENet  0.6658239 0.6929057 0.7021977 0.7122502 0.7243075 0.7853352    0
## PLS   0.6603439 0.6904538 0.7039741 0.7142321 0.7338252 0.7852501    0
## PCR   0.6556238 0.6728616 0.7075916 0.7125348 0.7416645 0.7908076    0
## GAM   0.6630045 0.6877732 0.7061953 0.7095810 0.7215971 0.7815306    0
## MARS  0.6728817 0.7037414 0.7127128 0.7165309 0.7265601 0.7861845    0
##
## Rsquared
##             Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.10452079 0.1598944 0.1845819 0.1861469 0.2324911 0.2596815    0
## Ridge 0.10758619 0.1639558 0.1858137 0.1876046 0.2267150 0.2605051    0
## Lasso 0.11036724 0.1635031 0.1934136 0.1911047 0.2337262 0.2707066    0
## ENet  0.10946131 0.1659847 0.1929010 0.1917926 0.2327736 0.2713318    0
## PLS   0.08935408 0.1624962 0.1872162 0.1866867 0.2231018 0.2520816    0
## PCR   0.12103550 0.1465347 0.1876839 0.1891519 0.2039132 0.3118517    0
## GAM   0.13096145 0.1549561 0.1991503 0.1967467 0.2297483 0.2800714    0
## MARS  0.12721372 0.1429318 0.1924540 0.1830814 0.2147103 0.2379142    0
```

```r
#visualize the comparison between cv rmse results
bwplot(res_comparison, metric = "RMSE", main = "CV RMSE Comparison")
```



5. Test RMSE

We additionally reported the test RMSE to describe how the selected model performed on the independent test set. Among the test RMSE results, the GAM also had the lowest value, indicating that it maintained the strongest predictive performance on unseen data. Although we are not using test RMSE for model selection, this result was consistent with the cross-validation findings and provided additional evidence that the GAM generalized well to the independent test set.

```
model_list <- list(LM = fit_lm, Ridge = fit_ridge, Lasso = fit_lasso,
                   ENet = fit_enet, PLS = fit_pls, PCR = fit_pcr,
                   GAM = fit_gam, MARS = fit_mars)


test_preds <- as.data.frame(lapply(model_list, predict, newdata = test_data))


test_preds$log_PFS <- test_data$log_PFS

# arrange the test rmse results into a table from ascending order
test_results <- test_preds |>
  pivot_longer(cols = -log_PFS, names_to = "Model", values_to = "Predicted")  |>
  group_by(Model)  |>
  summarize(
    Test_RMSE = sqrt(mean((log_PFS - Predicted)^2)),
```

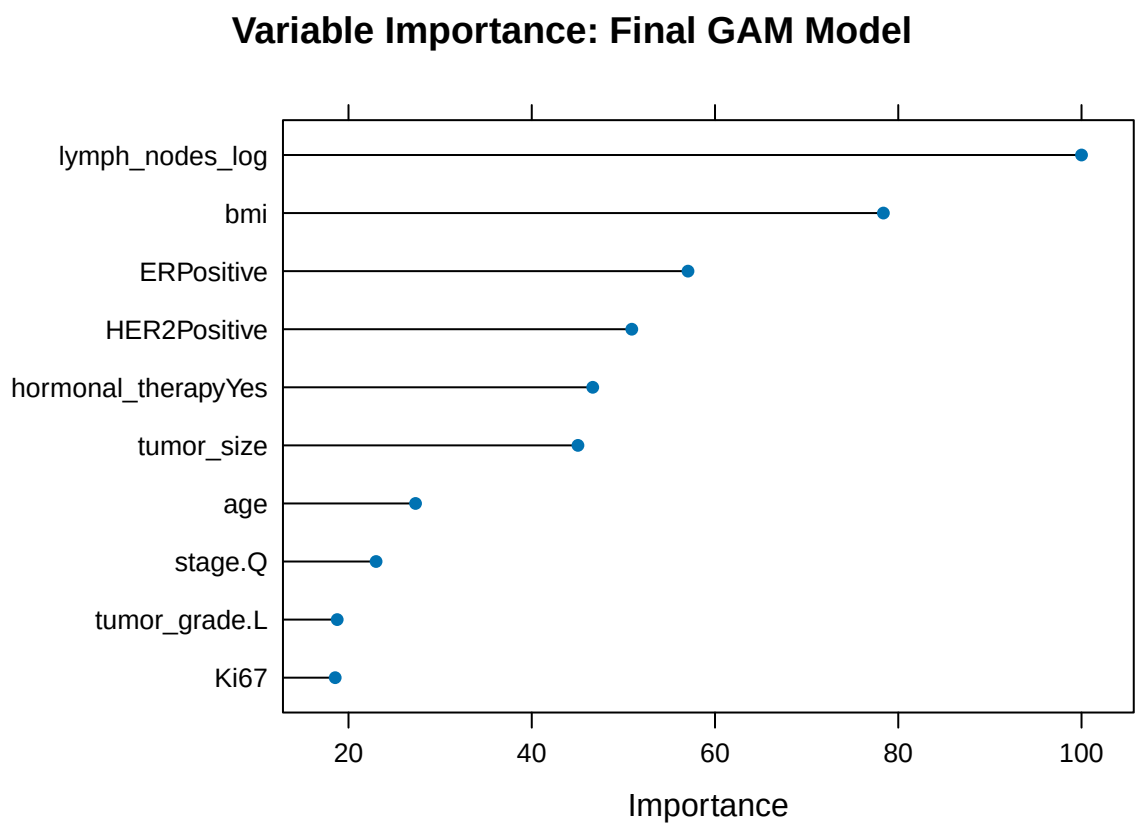
```r
  )  |>
  arrange(Test_RMSE)

print(test_results)
```

```
## # A tibble: 8 x 2
##   Model Test_RMSE
##   <chr>     <dbl>
## 1 GAM       0.686
## 2 PLS       0.687
## 3 Lasso     0.687
## 4 ENet      0.687
## 5 LM        0.689
## 6 Ridge     0.689
## 7 y         0.691
## 8 PCR       0.691
```

Variable Importance plot

Although the GAM output showed that many predictors had effects that were approximately linear (edf = 1.00), the model also suggested that some important predictors exhibited nonlinear patterns. To further justify the use of a nonlinear model, we examined the variable importance results from the GAM. We found that the two most influential predictors were both variables with nonlinear effects. This helps explain why the GAM achieved the best predictive performance.

```r
# Variable Importance for GAM model
gam_imp <- varImp(fit_gam)

plot(gam_imp, top = 10, main = "Variable Importance: Final GAM Model")
```



In conclusion, among all candidate models, the GAM showed the best predictive performance based on the lowest cross-validated RMSE. The test RMSE was reported as a supplementary measure on the independent test set, and it was consistent with the cross-validation results, with the GAM model also achieving the lowest value. Overall, for researcher A's data, log_PFS~menopause+race+tumor_grade+ER+PR+HER2 +stage+surgery+chemo+hormonal_therapy+ s(lymph_nodes_log)+s(tumor_size)+s(bmi)+s(Ki67)+s(age) from the GAM model is the best predictive model for PFS.

6. External validation - Researcher B

```
#load 2000-2005 cohort data
load("datB.RData")
dfB <- datB
```

Convert variables to factors

```
dfB <- dfB |>
  mutate(
    # Nominal variables
    menopause = factor(menopause, levels = c(0,1), labels = c("Pre","Post")),
    race = factor(race, levels = c(1,2,3,4),
                  labels = c("White","Black","Asian","Hispanic")),
```

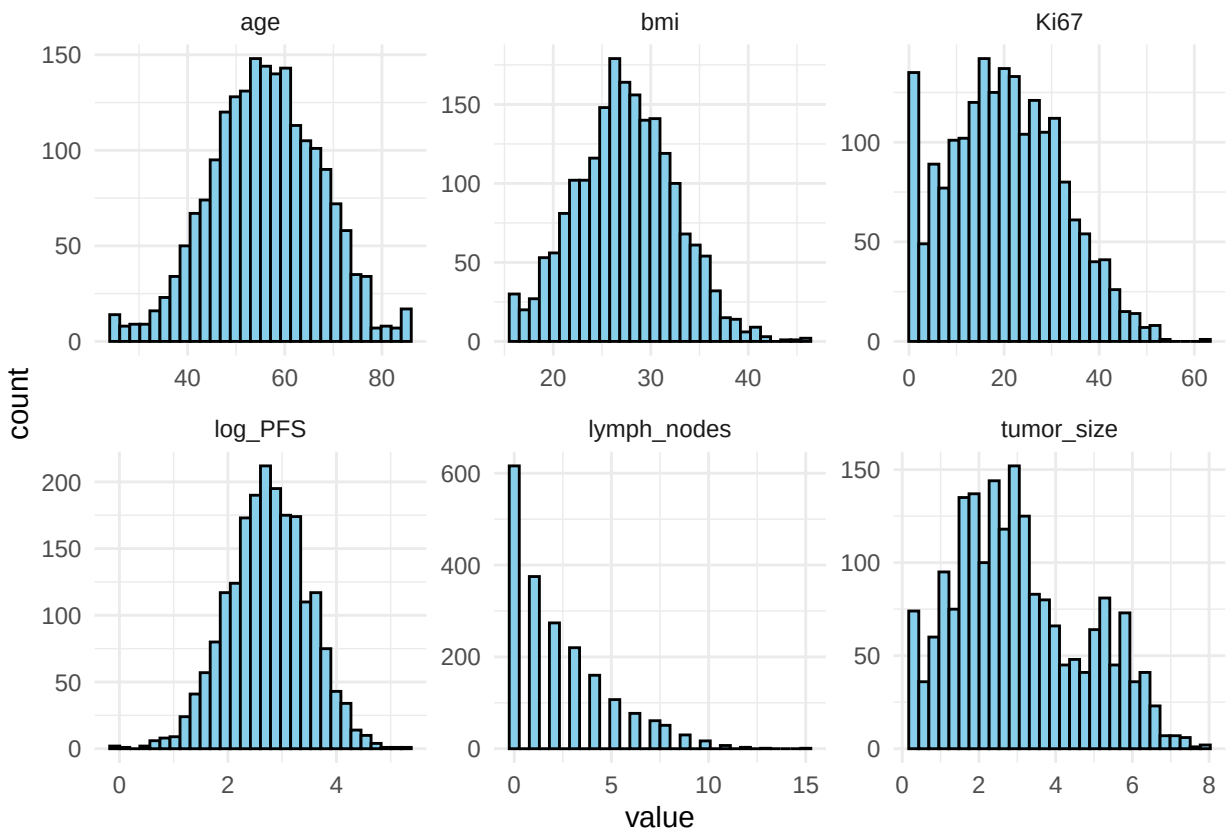
```r
    surgery = factor(surgery, levels = c(0,1),
                     labels = c("BCS","Mastectomy")),
    chemo = factor(chemo, levels = c(0,1),
                   labels = c("No","Yes")),
    hormonal_therapy = factor(hormonal_therapy, levels = c(0,1),
                              labels = c("No","Yes")),
    ER = factor(ER, levels = c(0,1), labels = c("Negative","Positive")),
    PR = factor(PR, levels = c(0,1), labels = c("Negative","Positive")),
    HER2 = factor(HER2, levels = c(0,1), labels = c("Negative","Positive")),

    # Ordinal variables
    tumor_grade = factor(tumor_grade, levels = c(1,2,3),
                         labels = c("Well","Moderate","Poor"),
                         ordered = TRUE),

    stage = factor(stage, levels = c(1,2,3),
                   labels = c("Stage I","Stage II","Stage III"),
                   ordered = TRUE)
  )
```

```r
summary(dfB)
```

```
##       id                 age        menopause         race           bmi
##  Length:2000        Min.   :25.00   Pre : 638   White   :1097   Min.   :16.00
##  Class :character   1st Qu.:48.50   Post:1362   Black   : 369   1st Qu.:24.00
##  Mode  :character   Median :56.00               Asian   : 220   Median :27.30
##                     Mean   :56.05               Hispanic: 314   Mean   :27.44
##                     3rd Qu.:63.80                               3rd Qu.:30.82
##                     Max.   :85.00                               Max.   :45.90
##    tumor_size      tumor_grade   lymph_nodes            ER
##  Min.   :0.300   Well    :502   Min.   : 0.000   Negative: 652
##  1st Qu.:1.800   Moderate:839   1st Qu.: 0.000   Positive:1348
##  Median :2.800   Poor    :659   Median : 2.000
##  Mean   :3.075                  Mean   : 2.318
##  3rd Qu.:4.200                  3rd Qu.: 4.000
##  Max.   :7.900                  Max.   :15.000
##         PR             HER2           Ki67             stage
##  Negative: 991   Negative:1526   Min.   : 1.00   Stage I  :692
##  Positive:1009   Positive: 474   1st Qu.:11.70   Stage II :828
##                                  Median :19.85   Stage III:480
##                                  Mean   :20.37
##                                  3rd Qu.:28.80
##                                  Max.   :62.30
##        surgery     chemo      hormonal_therapy    log_PFS
##  BCS       :1117   No : 702   No : 861         Min.   :-0.0866
##  Mastectomy: 883   Yes:1298   Yes:1139         1st Qu.: 2.2536
##                                                Median : 2.7440
##                                                Mean   : 2.7424
##                                                3rd Qu.: 3.2487
##                                                Max.   : 5.2864
```

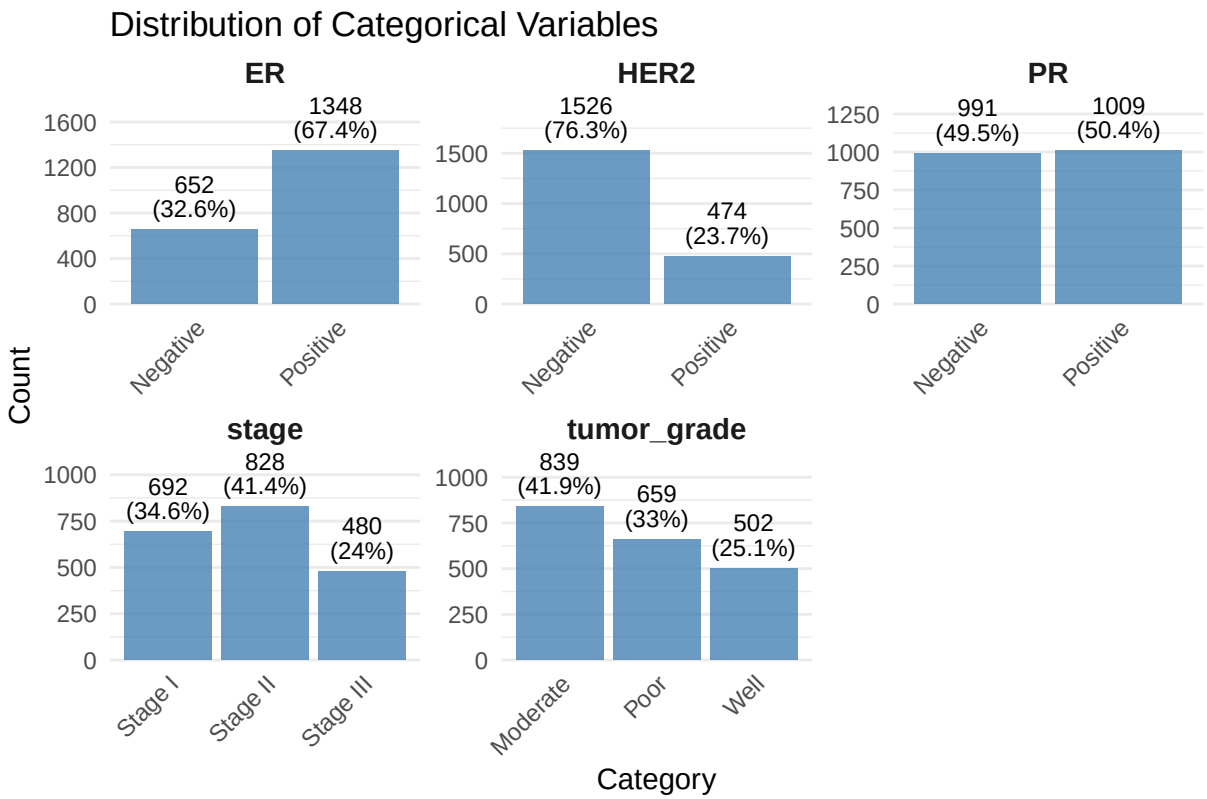
Distribution of continuous variables

```
# Histograms
dfB |>
  select(age, bmi, tumor_size, lymph_nodes, Ki67, log_PFS) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  facet_wrap(~name, scales = "free") +
  theme_minimal()
```

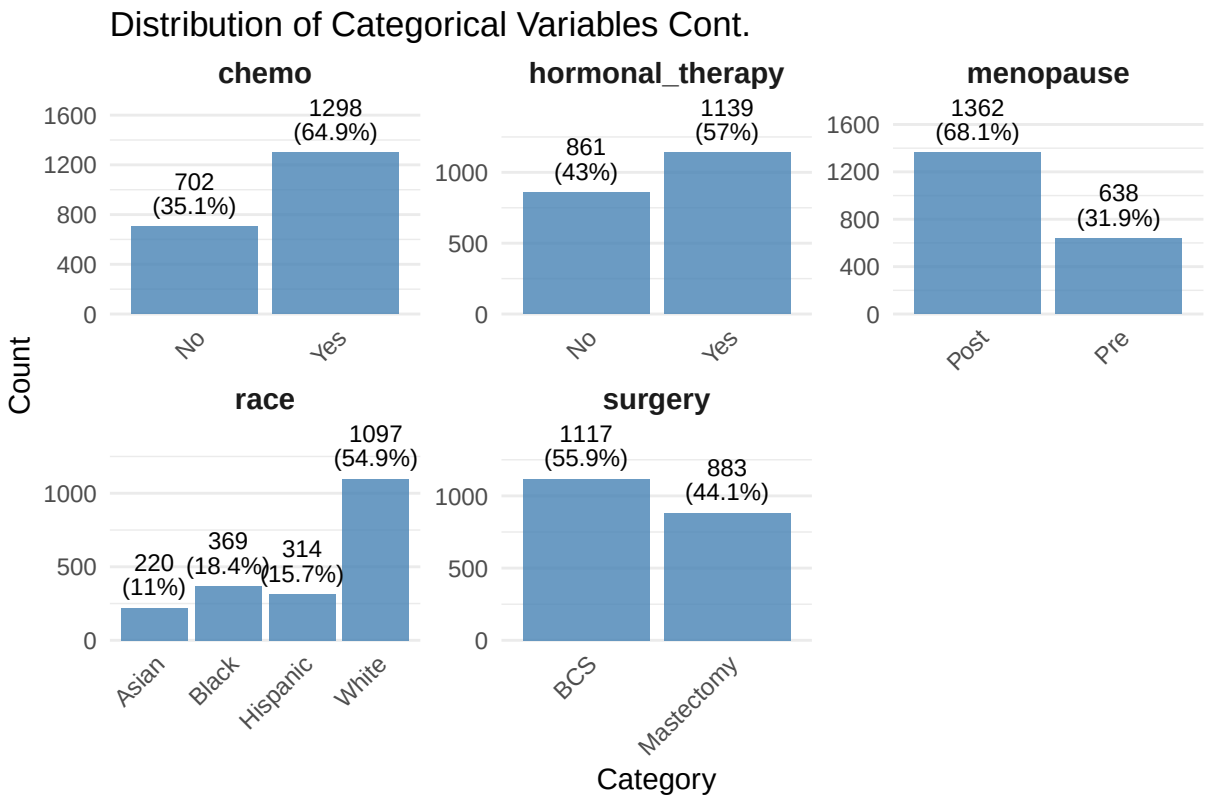


Frequency distribution of categorical variables

```
vars1 <- c("stage", "tumor_grade", "ER", "PR", "HER2")
vars2 <- c("race", "menopause", "chemo", "hormonal_therapy", "surgery")

plot_categorical(dfB, vars1, "Distribution of Categorical Variables")
```

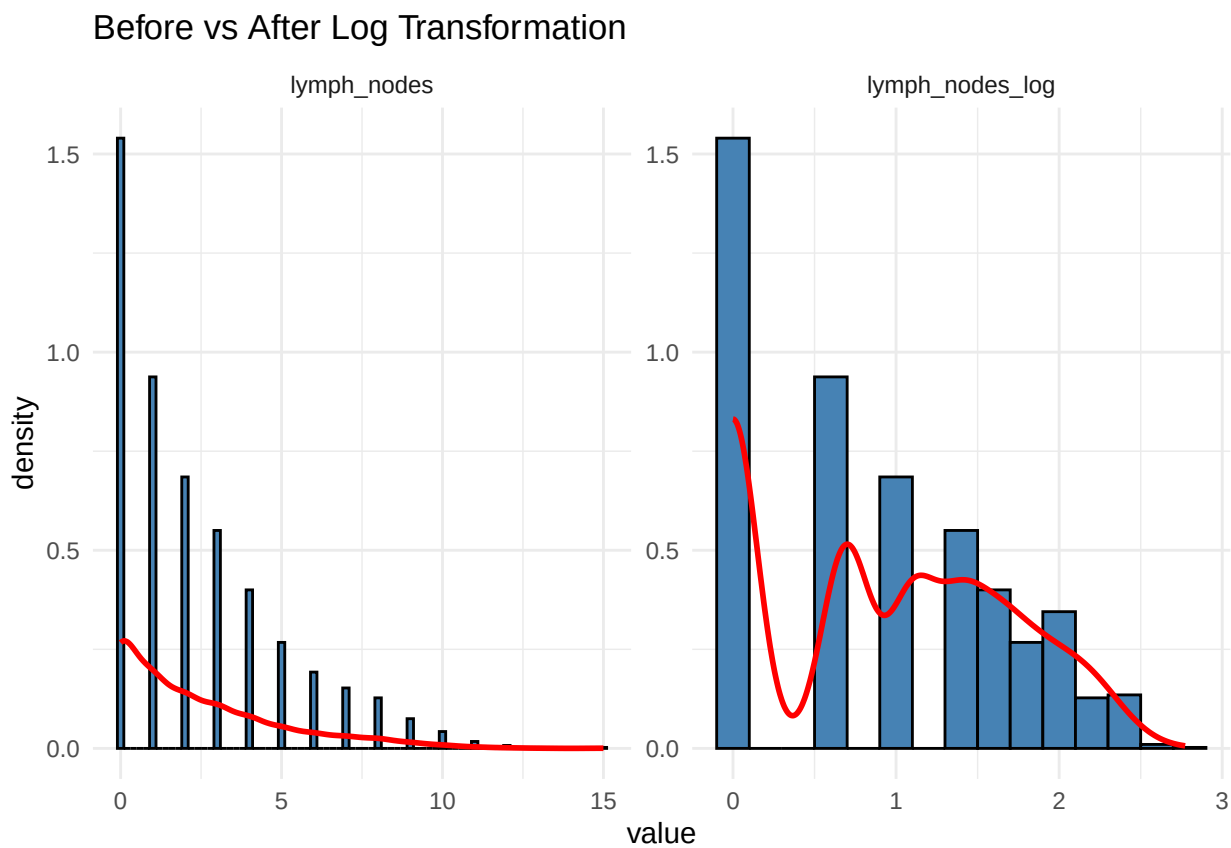



```r
plot_categorical(dfB, vars2, "Distribution of Categorical Variables Cont.")
```



Handle Skewness

```
# Log transform lymph_nodes
dfB <- dfB |>
  mutate(
    lymph_nodes_log = log(lymph_nodes + 1)
  )
dfB |>
  select(lymph_nodes, lymph_nodes_log) |>
  pivot_longer(cols = everything()) |>
  ggplot(aes(x = value)) +
  geom_histogram(aes(y = after_stat(density)),
                 binwidth = 0.2,
                 fill = "steelblue",
                 color = "black") +
  geom_density(color = "red", linewidth = 1) +
  facet_wrap(~name, scales = "free") +
  theme_minimal() +
  labs(title = "Before vs After Log Transformation")
```



Applying the GAM model to 2000-2005 cohort data

```
pred_B <- predict(fit_gam, newdata = dfB)
```

```
rmse_B <- sqrt(mean((datB$log_PFS - pred_B)^2))
print(rmse_B)
```

```
## [1] 0.8401839
```

The GAM model that we built to predict for PFS does not generalize well to the 2000–2005 cohort. The RMSE with 2000-2005 cohort data increased to 0.8401 compared to the original 0.686. This discrepancy suggests a temporal shift in patient outcomes.

7. Building New Model for 2000-2005 cohort data

we used the same 70/30 split methodology on the dataset.

```
set.seed(1) #to keep the same seed
train_index <- createDataPartition(dfB$log_PFS, p = 0.7, list = FALSE)
train_dataB <- dfB[train_index, ]
test_dataB  <- dfB[-train_index, ]
```

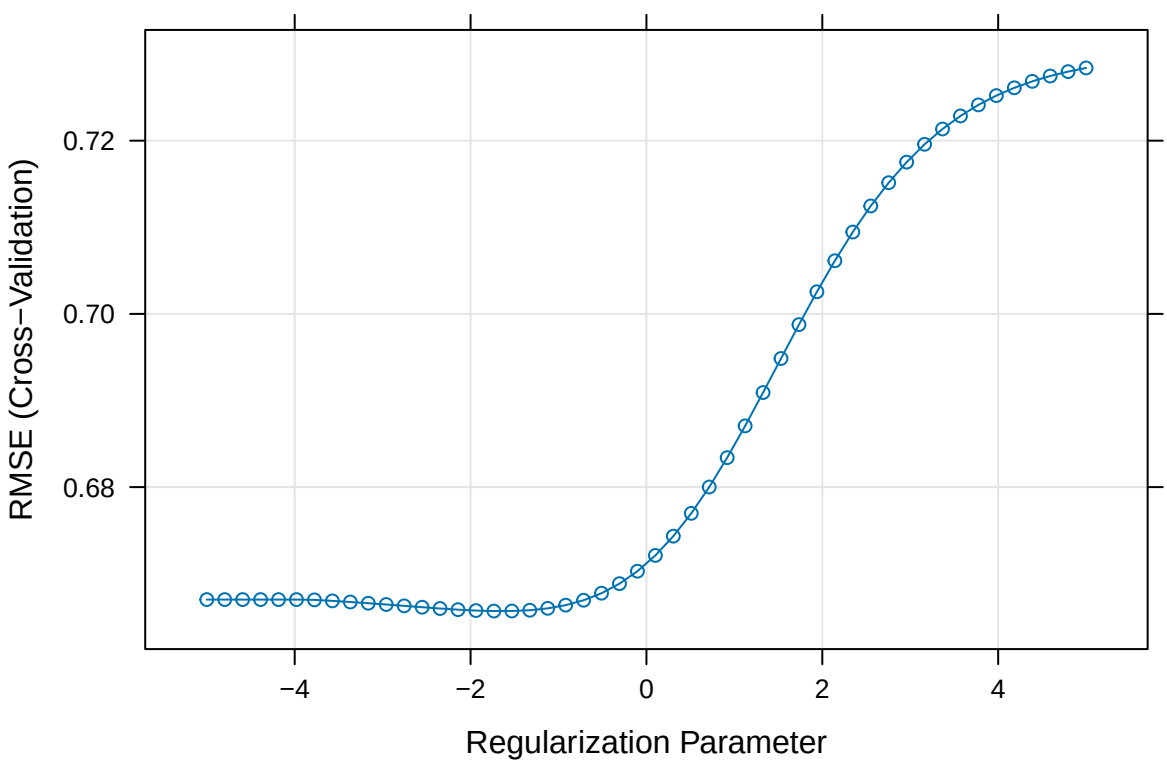
7-1. Linear Regression (Baseline)

```
set.seed(1)
fit_lmB <- train(full_formula, data = train_dataB, method = "lm", trControl = ctrl)
```

7-2. Ridge, Lasso, Elastic Net (Regularization)

7-2-1. Ridge Regression

```
set.seed(1)
fit_ridgeB <- train(full_formula, data = train_dataB, method = "glmnet",
                  tuneGrid = expand.grid(alpha = 0,
                                         lambda = exp(seq(-5, 5, length = 50))),
                  trControl = ctrl)

plot(fit_ridgeB, xTrans = log)
```




```
fit_ridgeB$bestTune
```

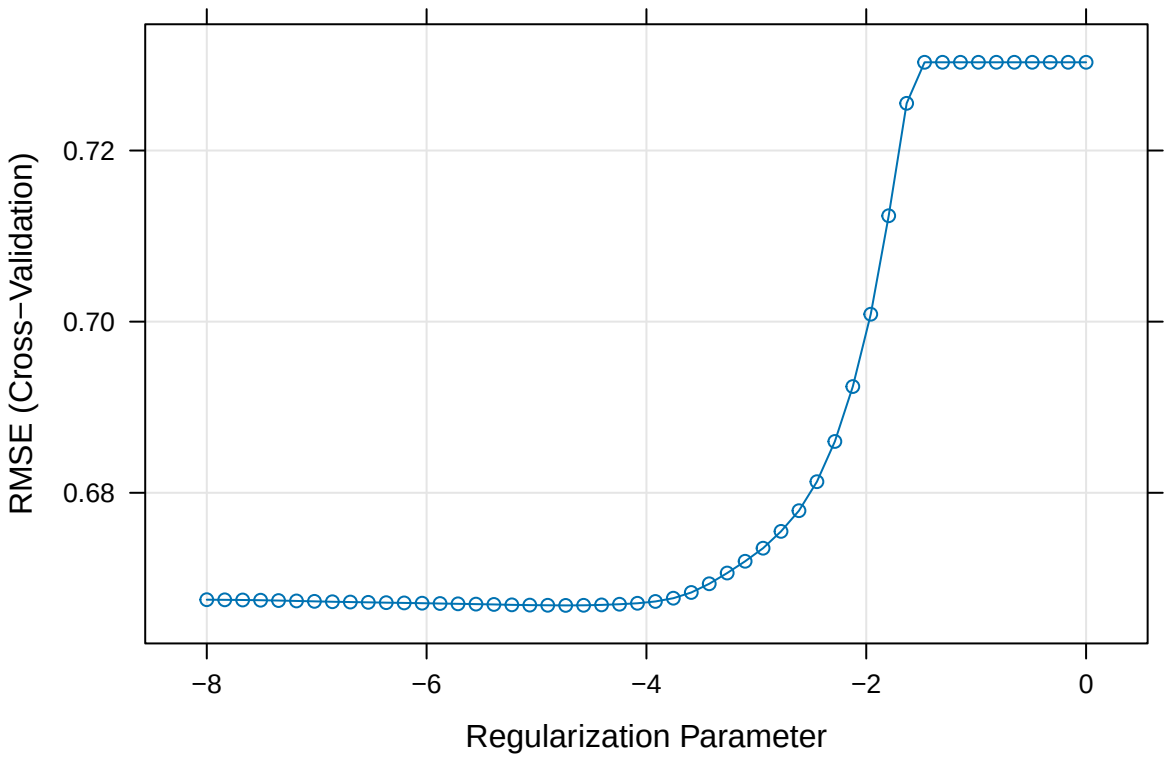
```
##    alpha    lambda
## 17     0 0.1764542
```

7-2-2. Lasso Regression

```
set.seed(1)
fit_lassoB <- train(full_formula, data = train_dataB, method = "glmnet",
                    tuneGrid = expand.grid(alpha = 1,
                                           lambda = exp(seq(-8, 0, length = 50))),
                    trControl = ctrl)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
plot(fit_lassoB, xTrans = log)
```




```
fit_lassoB$bestTune
```

```
##    alpha      lambda
## 21     1 0.008785138
```

```
coef(fit_lassoB$finalModel, fit_lassoB$bestTune$lambda)
```

```
## 20 x 1 sparse Matrix of class "dgCMatrix"
##                      s=0.008785138
```

```
## (Intercept)          3.2454032676
## age                 -0.0019214397
## menopausePost       -0.0323924132
## raceBlack           -0.0326891456
## raceAsian           -0.0093615433
## raceHispanic         0.1003616907
## bmi                 -0.0060419903
## tumor_size          -0.0745497353
## tumor_grade.L       -0.0009454298
## tumor_grade.Q       -0.0716599348
## lymph_nodes_log     -0.0284348697
## ERPositive           0.1801670343
## PRPositive           0.0094354740
## HER2Positive         0.0304294014
## Ki67                -0.0088934982
## stage.L             -0.0364155280
## stage.Q             -0.0001606731
## surgeryMastectomy    .
## chemoYes             .
## hormonal_therapyYes  0.1458624205
```

7-2-3. Elastic net Regression

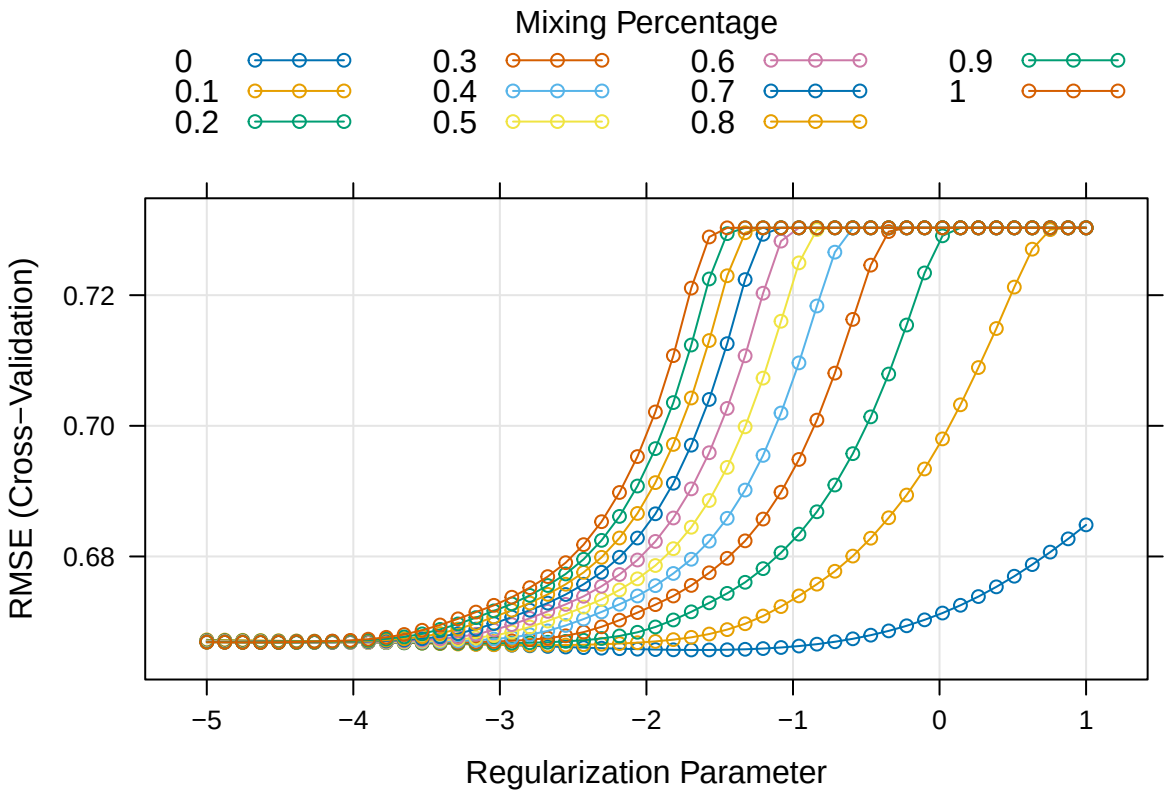
```r
set.seed(1)
fit_enetB <- train(full_formula, data = train_dataB, method = "glmnet",
                   tuneGrid = expand.grid(alpha = seq(0, 1, length = 11),
                                          lambda = exp(seq(-5, 1, length = 50))),
                   trControl = ctrl)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```r
fit_enetB$bestTune
```

```
##    alpha    lambda
## 28     0 0.1838054
```

```r
plot(fit_enetB, xTrans = log)
```




```r
coef(fit_enetB$finalModel, fit_enetB$bestTune$lambda)
```

```
## 20 x 1 sparse Matrix of class "dgCMatrix"
##                       s=0.1838054
## (Intercept)           3.166699667
## age                  -0.002074772
## menopausePost        -0.038837764
## raceBlack            -0.047730506
## raceAsian            -0.032963365
## raceHispanic          0.094047262
## bmi                  -0.006291346
## tumor_size           -0.046547631
## tumor_grade.L        -0.042427442
## tumor_grade.Q        -0.068967025
## lymph_nodes_log      -0.040422362
## ERPositive            0.158097688
## PRPositive            0.037974206
## HER2Positive          0.033621575
## Ki67                 -0.006728931
## stage.L              -0.083968381
## stage.Q              -0.020776636
## surgeryMastectomy    -0.017526687
## chemoYes             -0.013687916
## hormonal_therapyYes   0.142765775
```

7-3. PCR & PLS (Dimension Reduction)

7-3-1. PCR (Principal Component Regression)

```
set.seed(1)
pcr_modB <- pcr(full_formula, data = train_dataB, scale = TRUE, validation = "CV")

summary(pcr_modB)
```

```
## Data:    X dimension: 1400 19
##  Y dimension: 1400 1
## Fit method: svdpc
## Number of components considered: 19
##
## VALIDATION: RMSEP
## Cross-validated using 10 random segments.
##        (Intercept)  1 comps  2 comps  3 comps  4 comps  5 comps  6 comps
## CV          0.7311   0.6776   0.6700   0.6689   0.6689   0.6692   0.6688
## adjCV       0.7311   0.6776   0.6699   0.6688   0.6687   0.6691   0.6686
##        7 comps  8 comps  9 comps  10 comps  11 comps  12 comps  13 comps
## CV      0.6677   0.6677   0.6678    0.6685    0.6681    0.6685    0.6689
## adjCV   0.6675   0.6674   0.6675    0.6682    0.6678    0.6682    0.6685
##        14 comps  15 comps  16 comps  17 comps  18 comps  19 comps
## CV       0.6672    0.6676    0.6681    0.6675    0.6684    0.6690
## adjCV    0.6669    0.6672    0.6677    0.6670    0.6679    0.6684
##
## TRAINING: % variance explained
##          1 comps  2 comps  3 comps  4 comps  5 comps  6 comps  7 comps  8 comps
## X          17.92    31.03    39.91    46.45    52.70    58.71    64.38    69.52
## log_PFS    14.10    16.17    16.61    16.82    16.82    17.20    17.45    17.68
##          9 comps  10 comps  11 comps  12 comps  13 comps  14 comps  15 comps
## X          74.31     79.01     83.33     87.55     90.99     94.00     95.65
## log_PFS    17.70     17.77     17.91     17.91     17.93     18.41     18.41
##          16 comps  17 comps  18 comps  19 comps
## X           97.22     98.54     99.52    100.00
## log_PFS     18.41     18.70     18.72     18.77
```

```
fit_pcrB <- train(full_formula, data = train_data, method = "pcr",
                  scale = TRUE, trControl = ctrl, tuneLength = 15)
```

7-3-2. PLS (Partial Least Squares)

```
set.seed(1)
fit_plsB <- train(full_formula, data = train_dataB, method = "pls",
                  scale = TRUE, trControl = ctrl, tuneLength = 15)
```

7-4. GAM & MARS (Nonlinear Methods)

7-4-1. GAM

```r
set.seed(1)
fit_gamB <- train(full_formula, data = train_dataB, method = "gam", trControl = ctrl)

summary(fit_gamB)
```

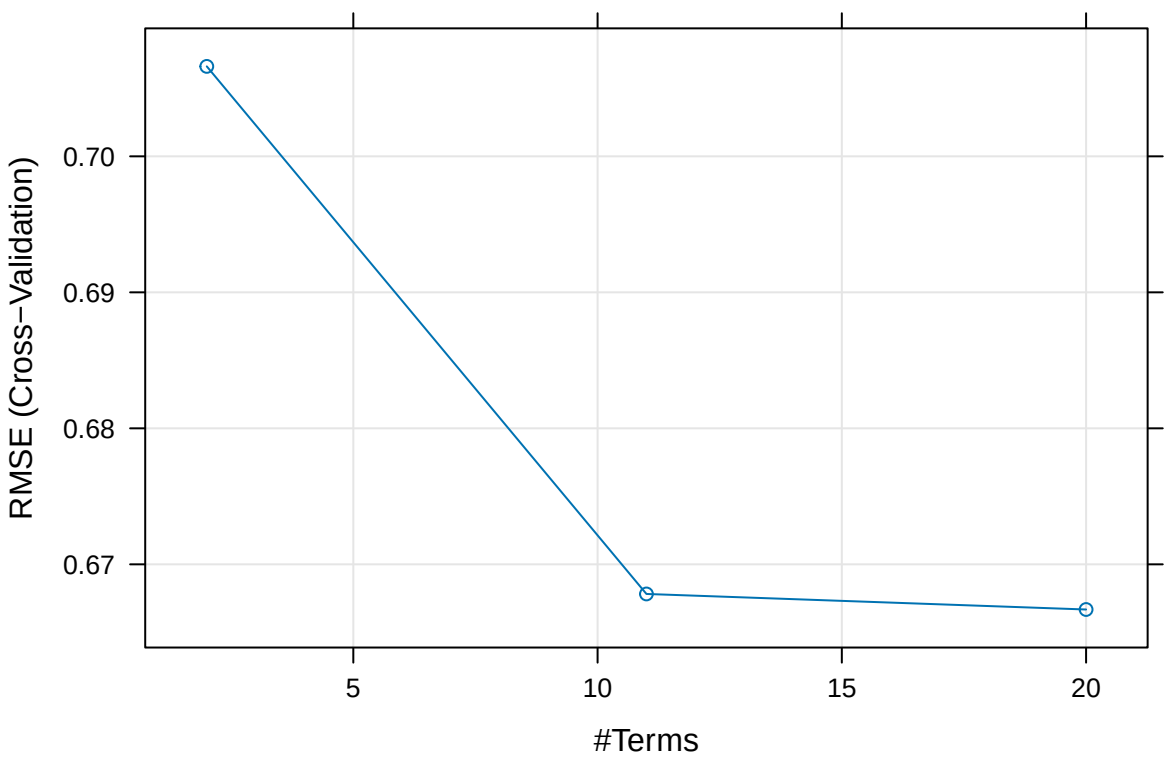
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## .outcome ~ menopausePost + raceBlack + raceAsian + raceHispanic +
##     tumor_grade.Q + ERPositive + PRPositive + HER2Positive +
##     stage.Q + surgeryMastectomy + chemoYes + hormonal_therapyYes +
##     tumor_grade.L + stage.L + s(lymph_nodes_log) + s(tumor_size) +
##     s(bmi) + s(Ki67) + s(age)
##
## Parametric coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2.5189723  0.0635483  39.639  < 2e-16 ***
## menopausePost       -0.0001449  0.0547032  -0.003  0.99789
## raceBlack           -0.0639260  0.0474836  -1.346  0.17844
## raceAsian           -0.0507336  0.0577231  -0.879  0.37960
## raceHispanic         0.1193044  0.0519641   2.296  0.02183 *
## tumor_grade.Q       -0.0741837  0.0295921  -2.507  0.01230 *
## ERPositive           0.2011271  0.0678442   2.965  0.00308 **
## PRPositive           0.0103256  0.0428114   0.241  0.80945
## HER2Positive         0.0466578  0.0423343   1.102  0.27060
## stage.Q              0.0328243  0.0486129   0.675  0.49965
## surgeryMastectomy   -0.0122215  0.0380349  -0.321  0.74801
## chemoYes            -0.0156760  0.0399008  -0.393  0.69447
## hormonal_therapyYes  0.1501901  0.0561329   2.676  0.00755 **
## tumor_grade.L       -0.0107207  0.0490312  -0.219  0.82696
## stage.L             -0.0206042  0.0952020  -0.216  0.82869
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                      edf Ref.df      F  p-value
## s(lymph_nodes_log) 2.743  3.370  2.382 0.083735 .
## s(tumor_size)      4.441  5.521  3.685 0.001895 **
## s(bmi)             3.249  4.114  6.351 4.29e-05 ***
## s(Ki67)            1.000  1.000 12.857 0.000348 ***
## s(age)             2.968  3.775  1.247 0.261954
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.201   Deviance explained = 21.7%
## GCV = 0.43586  Scale est. = 0.42671   n = 1400
```

```r
fit_gam$finalModel
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## .outcome ~ menopausePost + raceBlack + raceAsian + raceHispanic +
##     tumor_grade.Q + ERPositive + PRPositive + HER2Positive +
##     stage.Q + surgeryMastectomy + chemoYes + hormonal_therapyYes +
##     tumor_grade.L + stage.L + s(lymph_nodes_log) + s(tumor_size) +
##     s(bmi) + s(Ki67) + s(age)
##
## Estimated degrees of freedom:
## 2.24 2.25 3.50 1.00 1.00  total = 24.99
##
## GCV score: 0.4990726
```

7-4-2. MARS

```r
set.seed(1)
fit_marsB <- train(full_formula, data = train_dataB, method = "earth", trControl = ctrl)

plot(fit_marsB)
```

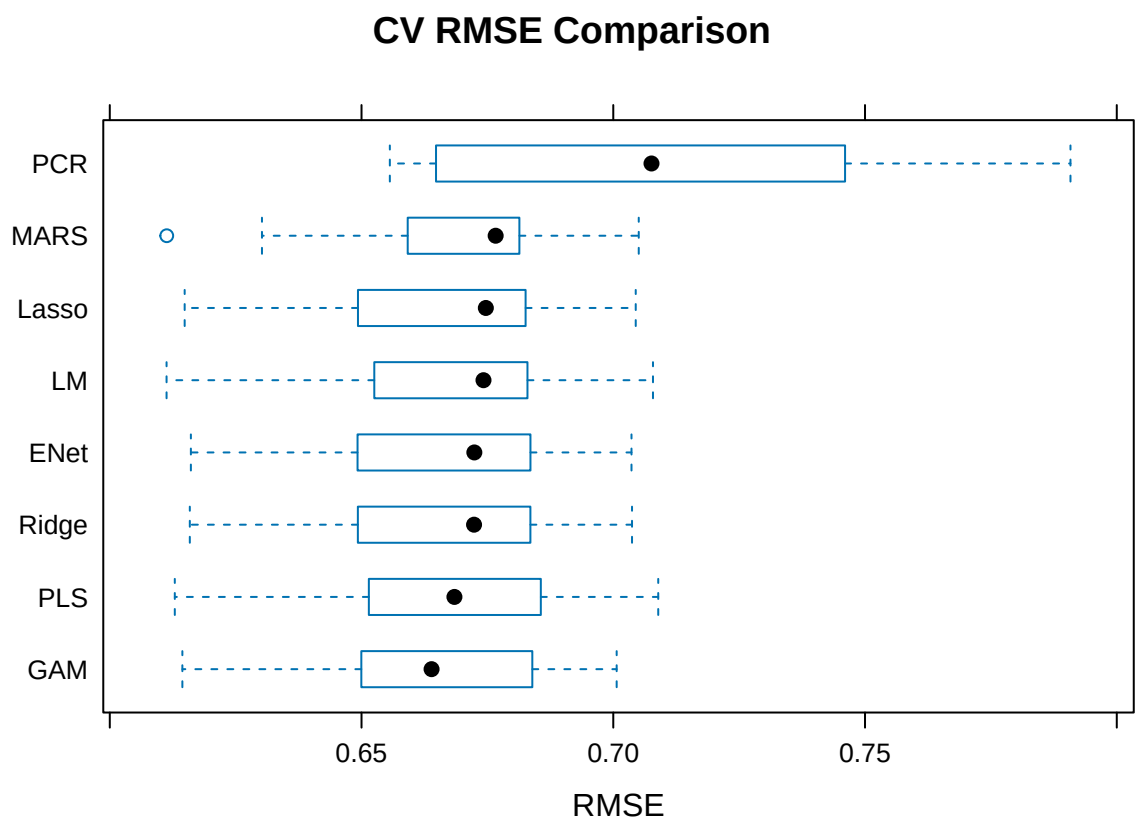


8. Model comparison CV RMSE

```
res_comparison <- resamples(list(LM = fit_lmB, Ridge = fit_ridgeB, Lasso = fit_lassoB, ENet = fit_enetB
summary(res_comparison)
```

```
##
## Call:
## summary.resamples(object = res_comparison)
##
## Models: LM, Ridge, Lasso, ENet, PLS, PCR, GAM, MARS
## Number of resamples: 10
##
## MAE
##            Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.4917313 0.5205963 0.5312504 0.5299579 0.5427277 0.5586178    0
## Ridge 0.4940739 0.5178695 0.5273754 0.5279860 0.5407840 0.5565413    0
## Lasso 0.4940525 0.5204421 0.5300700 0.5290305 0.5398371 0.5602214    0
## ENet  0.4942364 0.5178194 0.5273392 0.5279759 0.5407409 0.5565206    0
## PLS   0.4923366 0.5154731 0.5261544 0.5281519 0.5428808 0.5575514    0
## PCR   0.5304338 0.5394183 0.5616313 0.5663350 0.5850971 0.6179636    0
## GAM   0.4964719 0.5128098 0.5217428 0.5267049 0.5353060 0.5758156    0
## MARS  0.4965759 0.5171480 0.5304535 0.5293158 0.5403416 0.5753153    0
##
## RMSE
##            Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.6112744 0.6532413 0.6742207 0.6676874 0.6827245 0.7078882    0
## Ridge 0.6158954 0.6495321 0.6723495 0.6656874 0.6826663 0.7037252    0
## Lasso 0.6148743 0.6506043 0.6746869 0.6668390 0.6818752 0.7044595    0
## ENet  0.6161074 0.6494519 0.6724048 0.6656828 0.6826576 0.7036151    0
## PLS   0.6129114 0.6518700 0.6684491 0.6655355 0.6844692 0.7089318    0
## PCR   0.6556238 0.6728616 0.7075916 0.7125348 0.7416645 0.7908076    0
## GAM   0.6144142 0.6522544 0.6639084 0.6630735 0.6801593 0.7006712    0
## MARS  0.6113123 0.6607954 0.6766372 0.6666785 0.6804866 0.7050563    0
##
## Rsquared
##             Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## LM    0.08983162 0.1353493 0.1555753 0.1695309 0.2011536 0.3116426    0
## Ridge 0.09372381 0.1376696 0.1597629 0.1744204 0.2113836 0.3128093    0
## Lasso 0.08440242 0.1341214 0.1599644 0.1714751 0.2064124 0.3115982    0
## ENet  0.09378348 0.1377158 0.1597089 0.1744893 0.2114832 0.3127717    0
## PLS   0.09385905 0.1370589 0.1555123 0.1745541 0.2170402 0.3093812    0
## PCR   0.12103550 0.1465347 0.1876839 0.1891519 0.2039132 0.3118517    0
## GAM   0.05876708 0.1429085 0.1770441 0.1835747 0.2346718 0.2955786    0
## MARS  0.05209697 0.1412986 0.1841492 0.1760163 0.2046633 0.3085363    0
```

```
bwplot(res_comparison, metric = "RMSE", main = "CV RMSE Comparison")
```



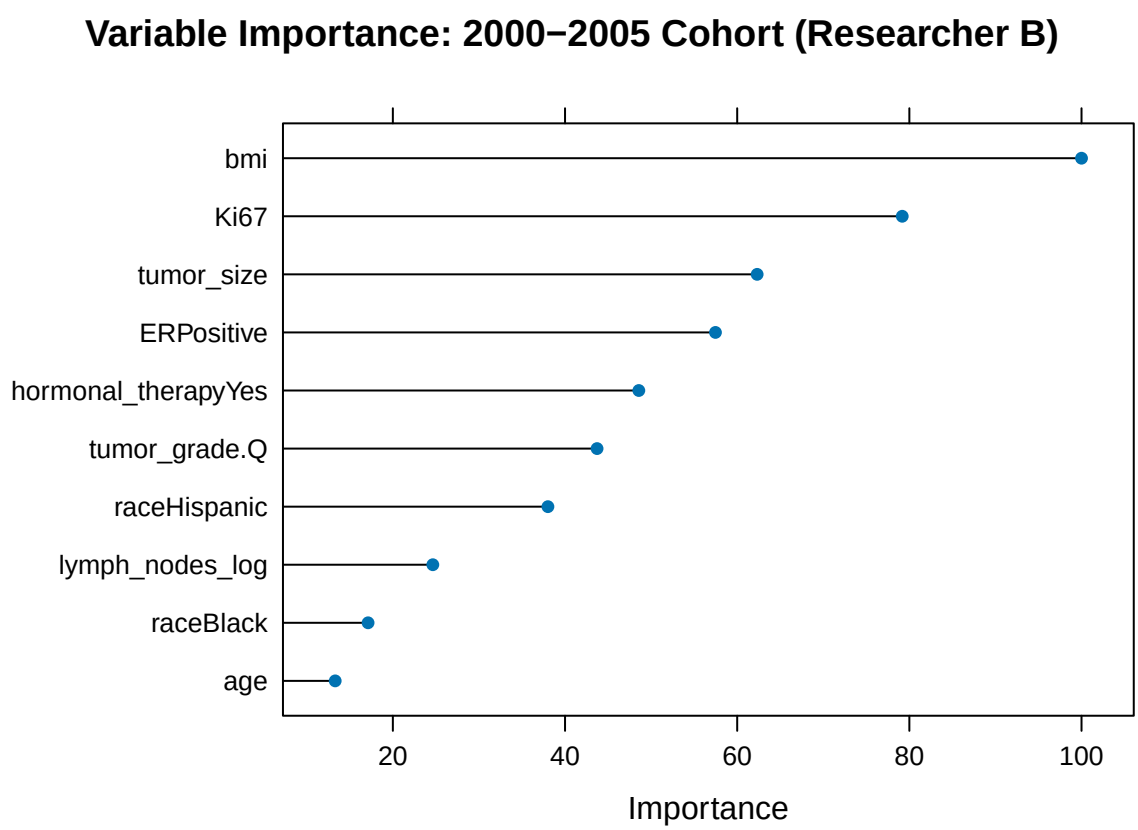
9.Test RMSE

```
model_list <- list(LM = fit_lmB, Ridge = fit_ridgeB, Lasso = fit_lassoB,
                   ENet = fit_enetB, PLS = fit_plsB, PCR = fit_pcrB,
                   GAM = fit_gamB, MARS = fit_marsB)


test_preds <- as.data.frame(lapply(model_list, predict, newdata = test_data))


test_preds$log_PFS <- test_data$log_PFS


test_results <- test_preds |>
  pivot_longer(cols = -log_PFS, names_to = "Model", values_to = "Predicted")  |>
  group_by(Model)  |>
  summarize(
    Test_RMSE = sqrt(mean((log_PFS - Predicted)^2)),
  )  |>
  arrange(Test_RMSE)

print(test_results)
```

```
## # A tibble: 8 x 2
##   Model Test_RMSE
##   <chr>     <dbl>
## 1 PCR       0.691
## 2 y         0.874
## 3 Lasso     0.877
## 4 PLS       0.878
## 5 Ridge     0.878
## 6 ENet      0.878
## 7 LM        0.880
## 8 GAM       0.881
```

In order to check what is the reason behind the discrepancy, we created a variable importance plot using GAM model that was trained with 2000-2005 cohort.

```r
#Variable Importance
vip_B <- varImp(fit_gamB)
plot(vip_B, top = 10, main = "Variable Importance: 2000-2005 Cohort (Researcher B)")
```



In the 1990-1995 cohort, lymph_nodes_log was the predominant predictor, whereas in the 2000-2005 cohort, its importance markedly decreased, giving way to treatment-related factors like hormonal_therapy and biological markers like Ki67 and tumor_grade.